



UE24MA241B — Linear Algebra & Its Applications

COMPLETE MASTER STUDY GUIDE: Units 3 & 4

Zero to Full Mastery | Exam-Ready | All Concepts + All Numericals

Course: UE24MA241B — Linear Algebra and Its Applications

Reference: Gilbert Strang, *Linear Algebra and Its Applications*, 5th Edition

Scope: Unit 3 (Orthogonality, Eigenvalues) + Unit 4 (SVD, Positive Definite, PCA)

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How to use this guide:

Read each section top-to-bottom the first time. For revision, jump to the Formula Sheets and Master Revision section. For practice, go directly to the Practice sections and attempt

problems before checking solutions.

UNIT 3: ORTHOGONALIZATION, EIGENVALUES & EIGENVECTORS

U3-1. Dot Product, Norm, and Angle

1.1 What is the Dot Product?

The **dot product** (also called the **inner product**) of two vectors $x, y \in \mathbb{R}^n$ is defined as:

$$x \cdot y = x^T y = \sum_{i=1}^n x_i y_i = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

Why it matters: The dot product measures how much two vectors "align" with each other. It is the foundation of orthogonality, projections, and least squares.

Properties (memorize all):

Property	Formula
Commutative	$x^T y = y^T x$
Distributive	$x^T (y + z) = x^T y + x^T z$
Scalar pull-out	$(cx)^T y = c(x^T y)$
Self dot product	$x^T x = x ^2 \geq 0$
Zero iff zero vector	$x^T x = 0 \iff x = 0$

1.2 Norm (Length) of a Vector

$$\|x\| = \sqrt{x^T x} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Unit vector: A vector u with $\|u\| = 1$.

To normalize any nonzero vector x : $u = \frac{x}{\|x\|}$

Worked Example 1.1:

Find the norm of $x = (3, -4, 0)$.

$$\|x\| = \sqrt{9 + 16 + 0} = \sqrt{25} = 5$$

Unit vector: $u = \frac{1}{5}(3, -4, 0)$

1.3 Angle Between Two Vectors

For nonzero vectors $x, y \in \mathbb{R}^n$:

$$\cos \theta = \frac{x^T y}{\|x\| \cdot \|y\|}$$

Key cases:

Angle θ	$\cos \theta$	Meaning
0°	1	Same direction
90°	0	Orthogonal (perpendicular)
180°	-1	Opposite direction

Orthogonality condition: $x \perp y \iff x^T y = 0$

1.4 Pythagorean Theorem and Law of Cosines

Starting from the expansion of $\|x - y\|^2$:

$$\|x - y\|^2 = (x - y)^T(x - y) = x^T x - 2x^T y + y^T y$$

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2x^T y$$

This is the **Law of Cosines** in vector form.

Special case — Pythagorean Theorem: If $x \perp y$ (so $x^T y = 0$):

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2 \quad (\text{when } x \perp y)$$

Worked Example 1.2:

Verify the Pythagorean theorem for $x = (1, 0)$ and $y = (0, 1)$.

$x^T y = 0$ (orthogonal). $\|x + y\|^2 = \|(1, 1)\|^2 = 2 = 1 + 1 = \|x\|^2 + \|y\|^2$. ✓

1.5 Transposes from the Inner Product

A key identity that connects A^T to dot products:

$$(Ax)^T y = x^T (A^T y)$$

Intuition: A^T is the matrix that "moves A across the dot product."

Worked Example 1.3:

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$, $x = (3, 1)$, $y = (2, 5)$.

- $Ax = (5, -1)$, so $(Ax)^T y = 10 - 5 = 5$
 - $A^T y = (2, -1)$, so $x^T (A^T y) = 6 - 1 = 5$ ✓
-

Common Mistakes:

- Forgetting that $\|x\|^2 = x^T x$, not $x \cdot x^T$ (which is a matrix)
 - Confusing $x^T y$ (a scalar) with xy^T (a matrix)
-

U3-2. Orthogonal Vectors and Subspaces

2.1 Orthogonal Vectors

Definition: Two vectors $x, y \in \mathbb{R}^n$ are **orthogonal** if:

$$x^T y = 0$$

Notation: $x \perp y$

Geometric meaning: They meet at a right angle (90°).

2.2 Key Theorem: Orthogonal $\implies$ Linearly Independent

Theorem: If nonzero vectors $v_1, v_2, \dots, v_k$ are **mutually orthogonal** (i.e., $v_i^T v_j = 0$ for all $i \neq j$), then they are **linearly independent**.

Why this matters for exams: This is a frequently tested fact. It means you can immediately conclude independence from orthogonality — no row reduction needed.

Proof sketch (not required but useful to understand):

Suppose $c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0$.

Take the dot product of both sides with v_i :

$$c_i (v_i^T v_i) = 0 \implies c_i = 0 \text{ (since } v_i \neq 0\text{)}.$$

So all $c_i = 0$, proving independence.

2.3 Worked Examples

Example 2.1: Show $v_1 = (1, 2, -1)$ and $v_2 = (2, -1, 0)$ are orthogonal.

$$v_1^T v_2 = (1)(2) + (2)(-1) + (-1)(0) = 2 - 2 + 0 = 0 \checkmark$$

Example 2.2: Find a vector in $\mathbb{R}^3$ orthogonal to both $(1, 1, 0)$ and $(1, 0, 1)$.

Let $x = (x_1, x_2, x_3)$.

Condition 1: $x_1 + x_2 = 0 \Rightarrow x_2 = -x_1$

Condition 2: $x_1 + x_3 = 0 \Rightarrow x_3 = -x_1$

Set $x_1 = 1$: $x = (1, -1, -1)$

Check: $(1)(1) + (-1)(1) + (-1)(0) = 0 \checkmark$ and $(1)(1) + (-1)(0) + (-1)(1) = 0 \checkmark$

Shortcut: This is the **cross product** in $\mathbb{R}^3$:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = (1 - 0)\mathbf{i} - (1 - 0)\mathbf{j} + (0 - 1)\mathbf{k} = (1, -1, -1)$$

2.4 Orthogonal Subspaces

Definition: Two subspaces S and T of $\mathbb{R}^n$ are **orthogonal** if:

$$x^T y = 0 \quad \text{for every } x \in S \text{ and every } y \in T$$

Important: This means *every* vector in S is orthogonal to *every* vector in T .

ASCII Diagram:

```

      R^n
     /  \
    S    T
  (subspace) (subspace)
    |        |
  +-----+
  All pairs orthogonal
  
```

U3-3. Orthogonal Complement

3.1 Definition

For a subspace $V \subseteq \mathbb{R}^n$, the **orthogonal complement** is:

$$V^\perp = \{x \in \mathbb{R}^n : x^T v = 0 \text{ for all } v \in V\}$$

Read as: " V -perp" — all vectors perpendicular to everything in V .

Key properties:

- $V^\perp$ is itself a subspace
- $(V^\perp)^\perp = V$
- $V \cap V^\perp = \{0\}$

3.2 Dimension Relationship

$$\dim(V) + \dim(V^\perp) = n$$

where n is the dimension of the ambient space $\mathbb{R}^n$.

Example: If V is a plane (2D) in $\mathbb{R}^3$, then $V^\perp$ is a line (1D). $2 + 1 = 3$. ✓

3.3 How to Find $V^\perp$ (Exam Method)

If $V = \text{span}\{v_1, v_2, \dots, v_k\}$, then $x \in V^\perp$ means:

$$x^T v_i = 0 \quad \text{for each } i = 1, 2, \dots, k$$

This is a **homogeneous linear system** — solve it!

Worked Example 3.1:

Let $V = \text{span}\{(1, 1, 0), (1, 0, 1)\} \subseteq \mathbb{R}^3$. Find $V^\perp$.

Set up the system for $x = (x_1, x_2, x_3)$:

$$x_1 + x_2 = 0$$

$$x_1 + x_3 = 0$$

From these: $x_2 = -x_1$ and $x_3 = -x_1$.

$$V^\perp = \text{span}\{(1, -1, -1)\}$$

Check: $\dim(V) = 2$, $\dim(V^\perp) = 1$, and $2 + 1 = 3$. ✓

Worked Example 3.2 (Medium):

Find the orthogonal complement of $V = \text{span}\{(1, 0, 1, 0), (0, 1, 0, 1)\}$ in $\mathbb{R}^4$.

System for $x = (x_1, x_2, x_3, x_4)$:

$$x_1 + x_3 = 0 \Rightarrow x_3 = -x_1$$

$$x_2 + x_4 = 0 \Rightarrow x_4 = -x_2$$

Free variables: x_1, x_2 .

$$V^\perp = \text{span}\{(1, 0, -1, 0), (0, 1, 0, -1)\}$$

$\dim(V^\perp) = 2$, and $2 + 2 = 4$. ✓

U3-4. Fundamental Theorem of Orthogonality

4.1 The Four Fundamental Subspaces

For an $m \times n$ matrix A :

Subspace	Lives in	Definition
Column space $C(A)$	$\mathbb{R}^m$	All combinations of columns of A
Null space $N(A)$	$\mathbb{R}^n$	Solutions to $Ax = 0$
Row space $C(A^T)$	$\mathbb{R}^n$	All combinations of rows of A
Left null space $N(A^T)$	$\mathbb{R}^m$	Solutions to $A^T y = 0$

4.2 Fundamental Orthogonality Relations

$$\boxed{C(A) \perp N(A^T)} \quad (\text{both in } \mathbb{R}^m)$$

$$\boxed{C(A^T) \perp N(A)} \quad (\text{both in } \mathbb{R}^n)$$

In words:

- The column space is orthogonal to the left null space
- The row space is orthogonal to the null space

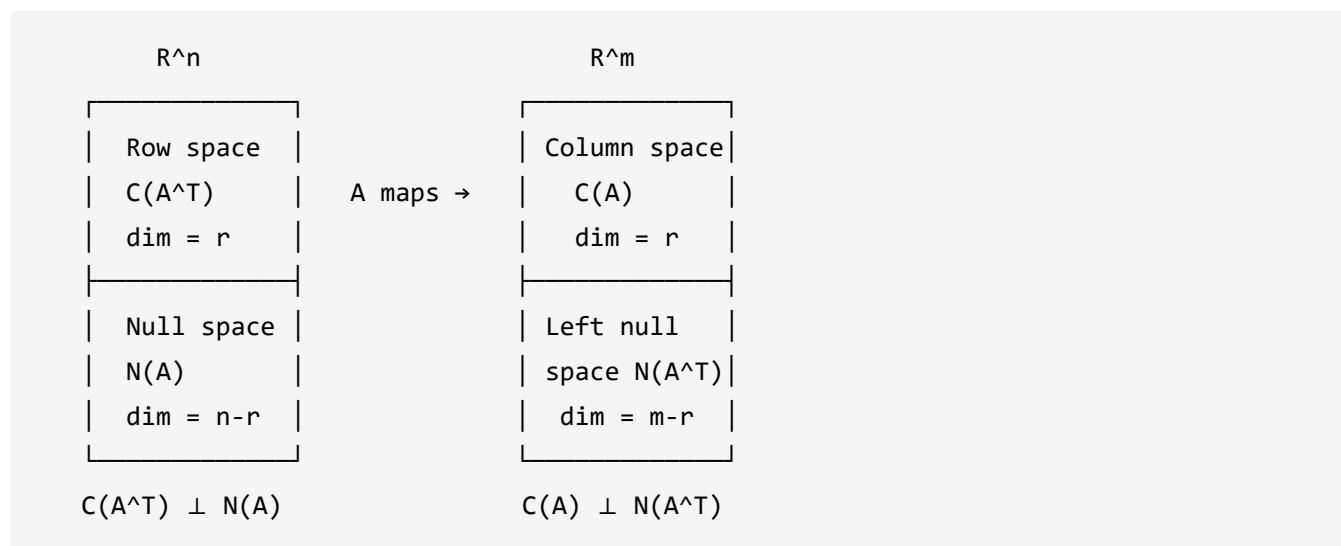
4.3 Dimension Relationships (Rank-Nullity)

If A is $m \times n$ with rank r :

$$\dim C(A) = r, \quad \dim N(A^T) = m - r$$

$$\dim C(A^T) = r, \quad \dim N(A) = n - r$$

4.4 Diagram



4.5 Worked Example (Verify the theorem)

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}$.

Columns: $a_1 = (1, 1, -2)$, $a_2 = (1, -1, 4)$.

A vector q is in $N(A^T)$ iff $A^T q = 0$, i.e., $a_1^T q = 0$ AND $a_2^T q = 0$.

This means q is orthogonal to every column of A — exactly the definition of $C(A) \perp N(A^T)$.

Common Exam Question: "Which subspaces are orthogonal to each other?"

Answer: Row space $\perp$ Null space; Column space $\perp$ Left null space.

U3-5. Projection onto a Line

5.1 Problem Setup

Given: A nonzero direction vector a (defining a line through the origin) and a vector b .

Goal: Find the point p on the line $\text{span}(a)$ that is **closest** to b .

Decomposition: $b = p + q$ where:

- $p \in \text{span}(a)$ — the **projection** (component along a)
- $q \perp a$ — the **error/residual** (component perpendicular to a)

5.2 Derivation of the Projection Formula

Since p lies on the line: $p = \hat{x} \cdot a$ for some scalar $\hat{x}$.

The condition $q \perp a$ means $a^T(b - p) = 0$:

$$a^T(b - \hat{x}a) = 0 \implies a^Tb - \hat{x}(a^Ta) = 0 \implies \hat{x} = \frac{a^Tb}{a^Ta}$$

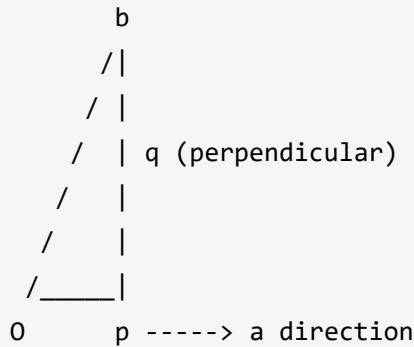
Therefore:

$$p = \text{proj}_a(b) = \frac{a^Tb}{a^Ta} \cdot a$$

and the residual:

$$q = b - p$$

5.3 Geometric Diagram



p is the "shadow" of b onto the line
 $q = b - p$ is perpendicular to a

5.4 Worked Examples

Example 5.1 (Basic):

Let $a = (1, 2, 2)$ and $b = (2, 1, 0)$. Find p and q .

$$a^T b = 1(2) + 2(1) + 2(0) = 4$$

$$a^T a = 1 + 4 + 4 = 9$$

$$p = \frac{4}{9}(1, 2, 2) = \left(\frac{4}{9}, \frac{8}{9}, \frac{8}{9}\right)$$

$$q = b - p = \left(2 - \frac{4}{9}, 1 - \frac{8}{9}, 0 - \frac{8}{9}\right) = \left(\frac{14}{9}, \frac{1}{9}, -\frac{8}{9}\right)$$

Verify: $a^T q = \frac{14}{9} + \frac{2}{9} - \frac{16}{9} = 0 \checkmark$

Example 5.2 (Medium):

Project $b = (1, 1, 1)$ onto the line spanned by $a = (1, 2, 3)$.

$$a^T b = 1 + 2 + 3 = 6, \quad a^T a = 1 + 4 + 9 = 14$$

$$p = \frac{6}{14}(1, 2, 3) = \frac{3}{7}(1, 2, 3) = \left(\frac{3}{7}, \frac{6}{7}, \frac{9}{7}\right)$$

$$q = \left(1 - \frac{3}{7}, 1 - \frac{6}{7}, 1 - \frac{9}{7}\right) = \left(\frac{4}{7}, \frac{1}{7}, -\frac{2}{7}\right)$$

Verify: $a^T q = \frac{4}{7} + \frac{2}{7} - \frac{6}{7} = 0 \checkmark$

Example 5.3 (Advanced — find the angle):

Find the angle between $b = (1, 2, 3)$ and its projection onto $a = (1, 0, 0)$.

Projection: $p = \frac{1}{1}(1, 0, 0) = (1, 0, 0)$.

$$\cos \theta = \frac{b^T p}{\|b\| \|p\|} = \frac{1}{\sqrt{14} \cdot 1} = \frac{1}{\sqrt{14}}$$

$$\theta = \arccos\left(\frac{1}{\sqrt{14}}\right) \approx 74.5^\circ$$

Common Mistakes:

- Writing $p = \frac{a^T b}{a^T b} \cdot a$ (wrong — denominator should be $a^T a$)
 - Forgetting to check $q \perp a$ as a verification step
-

U3-6. Projection Matrix (Rank 1)

6.1 The Projection Matrix onto a Line

The projection $p = \frac{a^T b}{a^T a} a$ can be written as a **matrix times** b :

$$p = \left(\frac{aa^T}{a^T a} \right) b = Pb$$

So the **rank-1 projection matrix** is:

$$P = \frac{aa^T}{a^T a}$$

Note: aa^T is an $n \times n$ matrix (outer product), while $a^T a$ is a scalar.

6.2 Properties of Projection Matrices

For any orthogonal projection matrix P :

Property	Formula	Why
Symmetric	$P^T = P$	$(aa^T)^T = aa^T$
Idempotent	$P^2 = P$	Projecting twice = projecting once
Eigenvalues	Only 0 or 1	$Px = x$ if $x \in C(P)$; $Px = 0$ if $x \perp C(P)$
Rank	$= \dim(\text{subspace})$	For line: rank 1

Proof of $P^2 = P$:

$$P^2 = \frac{aa^T}{a^T a} \cdot \frac{aa^T}{a^T a} = \frac{a(a^T a)a^T}{(a^T a)^2} = \frac{aa^T}{a^T a} = P$$

6.3 Worked Example

Let $a = (1, 2)$. Find the projection matrix P and verify $P^2 = P$.

$$aa^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad a^T a = 5$$

$$P = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Verify $P^2 = P$:

$$P^2 = \frac{1}{25} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}^2 = \frac{1}{25} \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = P \checkmark$$

U3-7. Projection onto a Subspace (Column Space)

7.1 Setup

Let A be an $m \times n$ matrix with **linearly independent columns**.

Goal: Find the vector $p \in C(A)$ closest to $b \in \mathbb{R}^m$.

Write $p = A\hat{x}$. The error $e = b - A\hat{x}$ must be orthogonal to the entire column space:

$$A^T(b - A\hat{x}) = 0$$

7.2 Normal Equations

$$\boxed{A^T A \hat{x} = A^T b}$$

These are the **normal equations**. When A has independent columns, $A^T A$ is invertible and:

$$\hat{x} = (A^T A)^{-1} A^T b$$

7.3 Projection and Projection Matrix

$$p = A\hat{x} = A(A^T A)^{-1} A^T b$$

So the **projection matrix onto $C(A)$** is:

$$P = A(A^T A)^{-1} A^T$$

The residual $q = b - p$ lies in $N(A^T)$ (the left null space).

7.4 Properties of $P = A(A^T A)^{-1} A^T$

- $P^T = P$ (symmetric)
 - $P^2 = P$ (idempotent)
 - $Pb = p$ (projection of b)
 - $(I - P)b = q$ (projection onto the orthogonal complement)
-

7.5 Full Worked Example (Exam-Level)

Problem: Find the projection of $b = (1, 2, 7)$ onto the column space of

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}$$

Split $b = p + q$ where $p \in C(A)$ and $q \in N(A^T)$.

Step 1: Compute $A^T A$.

Let $a_1 = (1, 1, -2)$ and $a_2 = (1, -1, 4)$.

$$a_1^T a_1 = 1 + 1 + 4 = 6$$

$$a_1^T a_2 = 1 - 1 - 8 = -8$$

$$a_2^T a_2 = 1 + 1 + 16 = 18$$

$$A^T A = \begin{bmatrix} 6 & -8 \\ -8 & 18 \end{bmatrix}$$

Step 2: Compute $A^T b$.

$$A^T b = \begin{bmatrix} a_1^T b \\ a_2^T b \end{bmatrix} = \begin{bmatrix} 1 + 2 - 14 \\ 1 - 2 + 28 \end{bmatrix} = \begin{bmatrix} -11 \\ 27 \end{bmatrix}$$

Step 3: Solve the normal equations.

$$\det(A^T A) = 6(18) - (-8)^2 = 108 - 64 = 44$$

$$(A^T A)^{-1} = \frac{1}{44} \begin{bmatrix} 18 & 8 \\ 8 & 6 \end{bmatrix}$$

$$\hat{x} = \frac{1}{44} \begin{bmatrix} 18 & 8 \\ 8 & 6 \end{bmatrix} \begin{bmatrix} -11 \\ 27 \end{bmatrix} = \frac{1}{44} \begin{bmatrix} -198 + 216 \\ -88 + 162 \end{bmatrix} = \frac{1}{44} \begin{bmatrix} 18 \\ 74 \end{bmatrix} = \begin{bmatrix} \frac{9}{22} \\ \frac{37}{22} \end{bmatrix}$$

Step 4: Compute $p = A\hat{x}$.

$$\begin{aligned} p &= \frac{9}{22}a_1 + \frac{37}{22}a_2 = \frac{1}{22} [9(1, 1, -2) + 37(1, -1, 4)] \\ &= \frac{1}{22}(9 + 37, 9 - 37, -18 + 148) = \frac{1}{22}(46, -28, 130) = \left(\frac{23}{11}, -\frac{14}{11}, \frac{65}{11} \right) \end{aligned}$$

Step 5: Compute $q = b - p$.

$$q = (1, 2, 7) - \left(\frac{23}{11}, -\frac{14}{11}, \frac{65}{11} \right) = \left(-\frac{12}{11}, \frac{36}{11}, \frac{12}{11} \right)$$

Verify $q \in N(A^T)$: $A^T q = 0$?

$$a_1^T q = -\frac{12}{11} + \frac{36}{11} - \frac{24}{11} = 0 \checkmark$$

$$a_2^T q = -\frac{12}{11} - \frac{36}{11} + \frac{48}{11} = 0 \checkmark$$

Final answer: $b = p + q$ where $p = \left(\frac{23}{11}, -\frac{14}{11}, \frac{65}{11} \right)$ and $q = \left(-\frac{12}{11}, \frac{36}{11}, \frac{12}{11} \right)$.

U3-8. Cauchy–Schwarz Inequality

8.1 Statement

For any vectors $a, b \in \mathbb{R}^n$:

$$|a^T b| \leq \|a\| \cdot \|b\|$$

Equality holds iff a and b are parallel (one is a scalar multiple of the other).

8.2 Why It Matters

- It guarantees $|\cos \theta| \leq 1$ (angles are well-defined)
- It is the foundation of the projection formula
- It appears in optimization and statistics

8.3 Triangle Inequality (consequence)

$$\|a + b\| \leq \|a\| + \|b\|$$

U3-9. Least Squares — Theory and Numericals

9.1 The Problem

When $Ax = b$ has **no exact solution** (overdetermined system: more equations than unknowns), we cannot satisfy all equations simultaneously.

Least squares solution: Find $\hat{x}$ that minimizes the **sum of squared errors**:

$$E(\hat{x}) = \|b - A\hat{x}\|^2 = \sum_{i=1}^m (b_i - (A\hat{x})_i)^2$$

9.2 Geometric Interpretation

The least squares solution finds the point $p = A\hat{x}$ in the column space $C(A)$ that is **closest** to b .

The error $e = b - A\hat{x}$ is **perpendicular to $C(A)$** .

b (actual)

/ |

/ |

/ | e = b - p (perpendicular to C(A))

/ |

/ |

p = $A\hat{x} \in C(A)$

$C(A)$ is the "floor"
p is the "shadow" of b on the floor
e is the "height" above the floor

9.3 Normal Equations (Derivation)

The condition $e \perp C(A)$ means $A^T e = 0$:

$$A^T(b - A\hat{x}) = 0 \implies \boxed{A^T A\hat{x} = A^T b}$$

When does a unique solution exist?

When A has **linearly independent columns** (i.e., $A^T A$ is invertible).

9.4 Summary of Least Squares

Quantity	Formula
Normal equations	$A^T A\hat{x} = A^T b$
Least squares solution	$\hat{x} = (A^T A)^{-1} A^T b$
Projection	$p = A\hat{x} = A(A^T A)^{-1} A^T b$
Residual	$e = b - p \in N(A^T)$

Quantity	Formula
Projection matrix	$P = A(A^T A)^{-1} A^T$

9.5 Worked Example (Overdetermined System)

Problem: Solve the overdetermined system in the least squares sense:

$$x_1 + x_2 = 1$$

$$x_1 - x_2 = 3$$

$$x_1 + 2x_2 = 2$$

$$\text{So } A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}.$$

$$\text{Step 1: } A^T A = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}, A^T b = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

(Check: $A^T b$ row 1 = $1 + 3 + 2 = 6$; row 2 = $1 - 3 + 4 = 2$)

$$\text{Step 2: } \det(A^T A) = 18 - 4 = 14$$

$$(A^T A)^{-1} = \frac{1}{14} \begin{bmatrix} 6 & -2 \\ -2 & 3 \end{bmatrix}$$

$$\text{Step 3: } \hat{x} = \frac{1}{14} \begin{bmatrix} 6 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 32 \\ -6 \end{bmatrix} = \begin{bmatrix} \frac{16}{7} \\ -\frac{3}{7} \end{bmatrix}$$

$$\text{Step 4: } p = A\hat{x} = \frac{16}{7}(1, 1, 1)^T - \frac{3}{7}(1, -1, 2)^T = \left(\frac{13}{7}, \frac{19}{7}, \frac{10}{7}\right)$$

$$e = b - p = \left(1 - \frac{13}{7}, 3 - \frac{19}{7}, 2 - \frac{10}{7}\right) = \left(-\frac{6}{7}, \frac{2}{7}, \frac{4}{7}\right)$$

Verify: $A^T e = 0$? $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} -6/7 \\ 2/7 \\ 4/7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \checkmark$

U3-10. Least Squares Line Fitting

10.1 Setup

Fit the best line $y = C + Dt$ to data points $(t_1, y_1), (t_2, y_2), \dots, (t_m, y_m)$.

Build the system $Ax = b$:

$$A = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

Normal equations give:

$$\begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum t_i y_i \end{bmatrix}$$

10.2 Worked Example (Classic)

Problem: Fit $y = C + Dt$ to the points $(0, 1), (1, 2), (2, 2), (3, 3)$.

Step 1: Build A and b .

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix}$$

Step 2: Compute $A^T A$ and $A^T b$.

$$A^T A = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 8 \\ 15 \end{bmatrix}$$

(Here: $\sum 1 = 4$, $\sum t = 6$, $\sum t^2 = 14$, $\sum y = 8$, $\sum ty = 15$)

Step 3: Solve.

$$\det = 4(14) - 36 = 20$$

$$(A^T A)^{-1} = \frac{1}{20} \begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix}$$

$$\begin{bmatrix} C \\ D \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} 8 \\ 15 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 112 - 90 \\ -48 + 60 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 22 \\ 12 \end{bmatrix} = \begin{bmatrix} 1.1 \\ 0.6 \end{bmatrix}$$

Best fit line: $y = 1.1 + 0.6t$

Step 4: Residuals.

$$e = b - A\hat{x} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1.1 \\ 1.7 \\ 2.3 \\ 2.9 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 0.3 \\ -0.3 \\ 0.1 \end{bmatrix}$$

Note: $\sum e_i = 0$ (always true for least squares with a constant term).

10.3 Worked Example (Fit a Parabola)

Problem: Fit $y = C + Dt + Et^2$ to $(0, 0), (1, 1), (2, 3), (3, 4)$.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 8 \\ 20 \\ 54 \end{bmatrix}$$

Solving (by elimination or inverse): $C \approx -0.05$, $D \approx 0.45$, $E \approx 0.25$.

Best fit parabola: $y \approx -0.05 + 0.45t + 0.25t^2$

U3-11. Weighted Least Squares

11.1 Motivation

Some measurements are more reliable than others. We assign **weights** $w_i > 0$ to each equation and minimize:

$$E_w = \sum_{i=1}^m w_i (b_i - (Ax)_i)^2$$

11.2 Weighted Normal Equations

Let $W = \text{diag}(w_1, w_2, \dots, w_m)$ be the diagonal weight matrix.

$$\boxed{A^T W A \hat{x} = A^T W b}$$

11.3 Worked Example

Problem: Fit $y = C + Dt$ to $(0, 1), (1, 2), (2, 3)$ with weights $w = (1, 2, 1)$.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^T W A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 6 \end{bmatrix}$$

$$A^T W b = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$$

Solving: $\det = 24 - 16 = 8$, so $\hat{x} = \frac{1}{8} \begin{bmatrix} 6 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 8 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Best fit line (weighted): $y = 1 + t$

U3-12. Orthonormal Vectors and Orthogonal Matrices

12.1 Orthonormal Vectors

Vectors $q_1, q_2, \dots, q_k$ are **orthonormal** if:

$$q_i^T q_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

In compact notation: $q_i^T q_j = \delta_{ij}$ (Kronecker delta).

Orthonormal = orthogonal + unit length.

12.2 Orthogonal Matrix

A **square** matrix Q is called **orthogonal** if its columns are orthonormal:

$$Q^T Q = I \implies Q^{-1} = Q^T$$

Note: If Q is square, $Q^T Q = I$ also implies $Q Q^T = I$.

What orthogonal matrices preserve:

- Length: $\|Qx\| = \|x\|$
- Dot products: $(Qx)^T (Qy) = x^T y$
- Angles between vectors
- Determinant: $\det(Q) = \pm 1$

Examples of orthogonal matrices:

- Rotation matrices
- Reflection matrices
- Permutation matrices
- Identity matrix

12.3 Rectangular Matrices with Orthonormal Columns

If Q is $m \times n$ with $m > n$ and orthonormal columns:

$$Q^T Q = I_n \quad (\text{but } QQ^T \neq I_m)$$

Why useful: Projection onto $C(Q)$ simplifies to $P = QQ^T$ (no inverse needed!).

12.4 Worked Example

Let $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

Verify $Q^T Q = I$:

$$Q^T Q = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I \checkmark$$

Find Q^{-1} : $Q^{-1} = Q^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

U3-13. Gram–Schmidt Orthogonalization Process

13.1 Goal

Convert a set of **linearly independent** vectors $\{v_1, v_2, \dots, v_k\}$ into an **orthonormal** set $\{q_1, q_2, \dots, q_k\}$ that spans the **same subspace**.

13.2 The Algorithm

Step 1: Start with $u_1 = v_1$.

Step 2: Subtract the projection of v_2 onto u_1 :

$$u_2 = v_2 - \frac{u_1^T v_2}{u_1^T u_1} u_1$$

Step 3: Subtract projections of v_3 onto u_1 and u_2 :

$$u_3 = v_3 - \frac{u_1^T v_3}{u_1^T u_1} u_1 - \frac{u_2^T v_3}{u_2^T u_2} u_2$$

General step k :

$$u_k = v_k - \sum_{j=1}^{k-1} \frac{u_j^T v_k}{u_j^T u_j} u_j$$

Normalize to get orthonormal vectors:

$$q_i = \frac{u_i}{\|u_i\|}$$

13.3 Key Idea (Intuition)

At each step, you remove the "shadow" of the new vector onto all previous orthogonal vectors. What remains is the part that is genuinely new and perpendicular to everything before.

```
v2
|  \
|   \ proj of v2 onto u1
|    \
|     \
|      \
u1-----> (direction of v1)
u2 = v2 - (projection) = perpendicular part
```

13.4 Worked Example 1 (in $\mathbb{R}^3$)

Problem: Apply Gram–Schmidt to $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$.

Step 1: $u_1 = v_1 = (1, 1, 0)$, $\|u_1\|^2 = 2$

Step 2:

$$u_1^T v_2 = 1, \quad \text{proj}_{u_1}(v_2) = \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

$$u_2 = (1, 0, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)$$

$$\|u_2\|^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$$

Step 3:

$$u_1^T v_3 = 1, \quad \text{proj}_{u_1}(v_3) = \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

$$u_2^T v_3 = \frac{1}{2}(0) + \left(-\frac{1}{2}\right)(1) + (1)(1) = \frac{1}{2}$$

$$\text{proj}_{u_2}(v_3) = \frac{1/2}{3/2}u_2 = \frac{1}{3} \left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}\right)$$

$$u_3 = (0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) - \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}\right) = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

Orthogonal basis: $\left\{ (1, 1, 0), \left(\frac{1}{2}, -\frac{1}{2}, 1\right), \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right) \right\}$

Normalize:

$$q_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad q_2 = \frac{1}{\sqrt{3/2}} \left(\frac{1}{2}, -\frac{1}{2}, 1 \right) = \sqrt{\frac{2}{3}} \left(\frac{1}{2}, -\frac{1}{2}, 1 \right)$$

$$q_3 = \frac{1}{\sqrt{4/3}} \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right) = \frac{1}{\sqrt{3}}(-1, 1, 1)$$

13.5 Worked Example 2 (in $\mathbb{R}^4$ — from your problem set)

Problem: Apply Gram–Schmidt to $v_1 = (1, 1, 1, 1)$, $v_2 = (1, 2, 4, 5)$, $v_3 = (1, -3, -4, -2)$.

Step 1: $u_1 = v_1 = (1, 1, 1, 1)$, $\|u_1\|^2 = 4$

Step 2:

$$u_1^T v_2 = 1 + 2 + 4 + 5 = 12$$

$$\text{proj}_{u_1}(v_2) = \frac{12}{4}(1, 1, 1, 1) = 3(1, 1, 1, 1) = (3, 3, 3, 3)$$

$$u_2 = (1, 2, 4, 5) - (3, 3, 3, 3) = (-2, -1, 1, 2)$$

$$\|u_2\|^2 = 4 + 1 + 1 + 4 = 10$$

Step 3:

$$u_1^T v_3 = 1 - 3 - 4 - 2 = -8$$

$$\text{proj}_{u_1}(v_3) = \frac{-8}{4}(1, 1, 1, 1) = -2(1, 1, 1, 1) = (-2, -2, -2, -2)$$

$$u_2^T v_3 = (-2)(1) + (-1)(-3) + (1)(-4) + (2)(-2) = -2 + 3 - 4 - 4 = -7$$

$$\text{proj}_{u_2}(v_3) = \frac{-7}{10}(-2, -1, 1, 2) = \left(\frac{7}{5}, \frac{7}{10}, -\frac{7}{10}, -\frac{7}{5}\right)$$

$$u_3 = v_3 - \text{proj}_{u_1}(v_3) - \text{proj}_{u_2}(v_3)$$

$$= (1, -3, -4, -2) - (-2, -2, -2, -2) - \left(\frac{7}{5}, \frac{7}{10}, -\frac{7}{10}, -\frac{7}{5}\right)$$

$$= (3, -1, -2, 0) - \left(\frac{7}{5}, \frac{7}{10}, -\frac{7}{10}, -\frac{7}{5}\right)$$

$$= \left(\frac{8}{5}, -\frac{17}{10}, -\frac{13}{10}, \frac{7}{5}\right)$$

Multiply by 10 to clear fractions: $w_3 = (16, -17, -13, 14)$

Orthogonal basis:

$$\boxed{\{(1, 1, 1, 1), (-2, -1, 1, 2), (16, -17, -13, 14)\}}$$

Orthonormal basis:

$$q_1 = \frac{1}{2}(1, 1, 1, 1), \quad q_2 = \frac{1}{\sqrt{10}}(-2, -1, 1, 2), \quad q_3 = \frac{1}{\sqrt{910}}(16, -17, -13, 14)$$

(where $\|(16, -17, -13, 14)\|^2 = 256 + 289 + 169 + 196 = 910$)

Common Mistakes in Gram–Schmidt:

- Forgetting to subtract ALL previous projections (not just the last one)
- Using v_j instead of u_j in the projection formula
- Not normalizing at the end when orthonormal basis is required

U3-14. QR Factorization

14.1 What is QR?

If A has linearly independent columns, Gram–Schmidt produces:

$$A = QR$$

where:

- Q has **orthonormal columns** ($Q^T Q = I$)
- R is **upper triangular** with positive diagonal entries

14.2 How to Build R

The entries of R come from the Gram–Schmidt projections:

$$r_{ij} = q_i^T v_j \quad (i \leq j)$$

The diagonal entries are the norms: $r_{ii} = \|u_i\|$.

14.3 Why QR is Useful

For least squares: $A = QR$ gives:

$$A^T A \hat{x} = A^T b \implies R^T Q^T Q R \hat{x} = R^T Q^T b \implies R \hat{x} = Q^T b$$

So: $\hat{x} = R^{-1} Q^T b$ (back-substitution on upper triangular R).

14.4 Worked Example

Problem: Find the QR factorization of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Step 1: $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$.

$$u_1 = v_1 = (1, 1, 0), \|u_1\| = \sqrt{2}, q_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$$

$$u_1^T v_2 = 1, \text{proj}_{u_1}(v_2) = \frac{1}{2}(1, 1, 0)$$

$$u_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = (\frac{1}{2}, -\frac{1}{2}, 1), \|u_2\| = \sqrt{3/2}$$

$$q_2 = \sqrt{\frac{2}{3}}(\frac{1}{2}, -\frac{1}{2}, 1) = \frac{1}{\sqrt{6}}(1, -1, 2)$$

Step 2: Build R .

$$r_{11} = \|u_1\| = \sqrt{2}, \quad r_{12} = q_1^T v_2 = \frac{1}{\sqrt{2}}, \quad r_{22} = \|u_2\| = \sqrt{3/2}$$

$$R = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} \end{bmatrix}$$

Verify: $A = QR$:

$$QR = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = A \checkmark$$

U3-15. Eigenvalues and Eigenvectors — Introduction

15.1 Why Do We Need Eigenvalues?

- Large matrices can be "compressed" to a few key numbers
- Eigenvectors are **directions that don't change** under a transformation (only scale)
- Eigenvalues tell you **how much** the scaling is
- Applications: stability analysis, quantum mechanics, Google PageRank, PCA, dynamical systems

15.2 Definition

A nonzero vector x is an **eigenvector** of matrix A with **eigenvalue** λ if:

$$Ax = \lambda x$$

Geometric meaning: Multiplying by A only stretches/shrinks x (by factor λ), without rotating it.

Before: x ----->

After: Ax -----> (same direction, scaled by λ)

If $\lambda > 1$: stretches

If $0 < \lambda < 1$: shrinks

If $\lambda < 0$: flips direction

If $\lambda = 0$: collapses to zero (A is singular)

15.3 How to Find Eigenvalues

Rearrange $Ax = \lambda x$ as $(A - \lambda I)x = 0$.

For a nonzero solution x to exist, the matrix $(A - \lambda I)$ must be **singular**:

$$\det(A - \lambda I) = 0$$

This is the **characteristic equation**. Solving it gives the eigenvalues.

15.4 How to Find Eigenvectors

For each eigenvalue λ_i , solve:

$$(A - \lambda_i I)x = 0$$

The solution space (null space of $A - \lambda_i I$) gives the eigenvectors.

15.5 Worked Example 1 (2×2 — Basic)

Problem: Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Step 1: Characteristic equation.

$$\det(A - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (2 - \lambda)^2 - 1 = 0$$

$$(2 - \lambda)^2 = 1 \implies 2 - \lambda = \pm 1$$

$$\lambda_1 = 3, \quad \lambda_2 = 1$$

Step 2: Eigenvectors.

For $\lambda_1 = 3$: $(A - 3I)x = 0$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0 \implies x_1 = x_2$$

Eigenvector: $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda_2 = 1$: $(A - I)x = 0$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0 \implies x_1 = -x_2$$

Eigenvector: $x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Verify: $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \checkmark$

15.6 Worked Example 2 (3×3 — Medium)

Problem: Find eigenvalues of $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 2 & -4 & 2 \end{bmatrix}$.

Characteristic equation:

$$\det(A - \lambda I) = \det \begin{bmatrix} 1 - \lambda & 2 & 0 \\ 0 & 3 - \lambda & 0 \\ 2 & -4 & 2 - \lambda \end{bmatrix}$$

Expanding along the third column (only one nonzero entry):

$$= (2 - \lambda) \det \begin{bmatrix} 1 - \lambda & 2 \\ 0 & 3 - \lambda \end{bmatrix} = (2 - \lambda)(1 - \lambda)(3 - \lambda)$$

Eigenvalues: $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$

Shortcut check: $\text{trace}(A) = 1 + 3 + 2 = 6 = 1 + 2 + 3 \checkmark, \det(A) = 6 = 1 \cdot 2 \cdot 3 \checkmark$

15.7 Worked Example 3 (Triangular Matrix — Shortcut)

Problem: Find eigenvalues of $A = \begin{bmatrix} 3 & 1 & 5 \\ 0 & 2 & 7 \\ 0 & 0 & 4 \end{bmatrix}$.

Shortcut: For triangular matrices, eigenvalues are the **diagonal entries**.

$$\lambda_1 = 3, \quad \lambda_2 = 2, \quad \lambda_3 = 4$$

Why: $\det(A - \lambda I) = (3 - \lambda)(2 - \lambda)(4 - \lambda)$ (product of diagonal entries of $A - \lambda I$).

15.8 Worked Example 4 (Repeated Eigenvalue)

Problem: Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.

Characteristic equation:

$$\det(A - \lambda I) = (3 - \lambda)^2 = 0 \implies \lambda = 3 \text{ (repeated)}$$

Eigenvectors for $\lambda = 3$:

$$(A - 3I)x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = 0 \implies x_2 = 0$$

Only one independent eigenvector: $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Note: This matrix is **not diagonalizable** (only 1 independent eigenvector for a 2×2 matrix).

U3-16. Properties of Eigenvalues and Eigenvectors

16.1 Core Properties (Memorize All)

Property	Statement
Trace = sum of eigenvalues	$\text{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$
Determinant = product of eigenvalues	$\det(A) = \lambda_1 \cdot \lambda_2 \cdots \lambda_n$
Triangular matrix	Eigenvalues are diagonal entries
Distinct eigenvalues	Eigenvectors are linearly independent
Symmetric matrix	Eigenvalues are real; eigenvectors for different λ are orthogonal
A is singular	$\lambda = 0$ is an eigenvalue
A is invertible	All eigenvalues $\neq 0$

16.2 Eigenvalues of Related Matrices

Matrix	Eigenvalues	Eigenvectors
A	λ	x

Matrix	Eigenvalues	Eigenvectors
A^{-1}	$1/\lambda$	x (same)
A^k	λ^k	x (same)
$A - cI$	$\lambda - c$	x (same)
cA	$c\lambda$	x (same)
A^T	λ (same values)	Different vectors

16.3 Properties of Eigenvectors

- Eigenvectors are **not unique** — any nonzero scalar multiple is also an eigenvector
- The **eigenspace** for λ is the null space of $(A - \lambda I)$
- Eigenvectors for **distinct** eigenvalues are **linearly independent**
- For **symmetric** matrices: eigenvectors for different eigenvalues are **orthogonal**

16.4 Worked Example (Using Properties)

Problem: If A has eigenvalues 2, 3, 5, find:

- $\det(A)$
- $\text{tr}(A)$
- Eigenvalues of A^{-1}
- Eigenvalues of $A^2 - 3I$

Solutions:

- $\det(A) = 2 \cdot 3 \cdot 5 = 30$
 - $\text{tr}(A) = 2 + 3 + 5 = 10$
 - Eigenvalues of A^{-1} : $\frac{1}{2}, \frac{1}{3}, \frac{1}{5}$
 - Eigenvalues of $A^2 - 3I$: $2^2 - 3 = 1, 3^2 - 3 = 6, 5^2 - 3 = 22$
-

16.5 Worked Example (Symmetric Matrix — Orthogonal Eigenvectors)

Problem: For $A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$, find eigenvalues and verify eigenvectors are orthogonal.

Characteristic equation:

$$\det(A - \lambda I) = (4 - \lambda)(1 - \lambda) - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5) = 0$$

$$\lambda_1 = 0, \lambda_2 = 5$$

Eigenvectors:

$$\text{For } \lambda_1 = 0: (A - 0)x = 0 \implies 4x_1 + 2x_2 = 0 \implies x_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\text{For } \lambda_2 = 5: (A - 5I)x = 0 \implies -x_1 + 2x_2 = 0 \implies x_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Verify orthogonality: $x_1^T x_2 = (1)(2) + (-2)(1) = 0$ ✓ (symmetric matrix property)

U3-17. Cayley–Hamilton Theorem

17.1 Statement

Cayley–Hamilton Theorem: Every square matrix A satisfies its own characteristic polynomial.

If $p(\lambda) = \det(\lambda I - A)$ is the characteristic polynomial, then:

$$p(A) = 0 \quad (\text{the zero matrix})$$

Example: If $p(\lambda) = \lambda^2 - 5\lambda + 6$, then $A^2 - 5A + 6I = 0$.

17.2 How to Use It

To find A^{-1} : If $p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$, then:

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1A + c_0I = 0$$

Multiply by A^{-1} (if $c_0 \neq 0$, i.e., A is invertible):

$$A^{-1} = -\frac{1}{c_0}(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1I)$$

To find high powers of A : Express A^k in terms of lower powers using the characteristic equation.

17.3 Worked Example 1 (2×2 — Find A^{-1})

Problem: For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, use Cayley–Hamilton to find A^{-1} .

Step 1: Characteristic polynomial.

$$p(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$$

Step 2: By Cayley–Hamilton: $A^2 - 4A + 3I = 0$.

Step 3: Multiply by A^{-1} :

$$A - 4I + 3A^{-1} = 0 \implies A^{-1} = \frac{1}{3}(4I - A)$$

$$A^{-1} = \frac{1}{3} \left(\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \right) = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Verify: $AA^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = I \checkmark$

17.4 Worked Example 2 (Find A^3 using Cayley–Hamilton)

Problem: For $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, find A^3 using Cayley–Hamilton.

Step 1: Characteristic polynomial.

$$p(\lambda) = (1 - \lambda)(3 - \lambda) = \lambda^2 - 4\lambda + 3$$

Step 2: By Cayley–Hamilton: $A^2 - 4A + 3I = 0 \implies A^2 = 4A - 3I$.

Step 3: Find $A^3 = A \cdot A^2 = A(4A - 3I) = 4A^2 - 3A = 4(4A - 3I) - 3A = 13A - 12I$.

$$A^3 = 13 \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} - 12 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}$$

Verify directly: $A^2 = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}$, $A^3 = \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}$ ✓

17.5 Worked Example 3 (3×3 Matrix)

Problem: Verify Cayley–Hamilton for $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

Characteristic polynomial: $p(\lambda) = (1 - \lambda)(2 - \lambda)(3 - \lambda) = -\lambda^3 + 6\lambda^2 - 11\lambda + 6$

Cayley–Hamilton: $-A^3 + 6A^2 - 11A + 6I = 0$

Since A is diagonal: $A^k = \text{diag}(1^k, 2^k, 3^k)$.

$$-A^3 + 6A^2 - 11A + 6I = \text{diag}(-1 + 6 - 11 + 6, -8 + 24 - 22 + 6, -27 + 54 - 33 + 6) =$$

U3-18. Diagonalization of a Matrix

18.1 What is Diagonalization?

A matrix A is **diagonalizable** if it can be written as:

$$A = S\Lambda S^{-1}$$

where:

- Λ is a **diagonal matrix** with eigenvalues on the diagonal
- S is an **invertible matrix** whose columns are the corresponding eigenvectors

18.2 When is A Diagonalizable?

Sufficient condition: A has n linearly independent eigenvectors.

Guaranteed cases:

- A has n **distinct** eigenvalues (always diagonalizable)
- A is **symmetric** (always diagonalizable, with orthogonal S)

Not diagonalizable: When there are repeated eigenvalues but not enough independent eigenvectors.

18.3 Steps to Diagonalize

1. Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$
2. Find eigenvectors $x_1, x_2, \dots, x_n$ (one for each eigenvalue)
3. Form $S = [x_1 | x_2 | \dots | x_n]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$
4. Then $A = S\Lambda S^{-1}$

18.4 Special Case: Symmetric Matrices

For symmetric A ($A^T = A$):

- Eigenvalues are real
- Eigenvectors for different eigenvalues are orthogonal
- A is **orthogonally diagonalizable**: $A = Q\Lambda Q^T$ where Q is orthogonal

$$A = Q\Lambda Q^T \quad (\text{symmetric matrices})$$

18.5 Power of Diagonalization

$$A^k = S \Lambda^k S^{-1} = S \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} S^{-1}$$

This makes computing high powers of A trivial!

18.6 Worked Example 1 (Full Diagonalization)

Problem: Diagonalize $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Step 1: Eigenvalues: $\lambda_1 = 3, \lambda_2 = 1$ (from Example 15.1).

Step 2: Eigenvectors: $x_1 = (1, 1), x_2 = (1, -1)$.

Step 3: Form S and Λ :

$$S = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Step 4: Find S^{-1} :

$$\det(S) = -2, \quad S^{-1} = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Verify: $S \Lambda S^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = A \checkmark$

Orthogonal diagonalization (since A is symmetric):

Normalize eigenvectors: $q_1 = \frac{1}{\sqrt{2}}(1, 1), q_2 = \frac{1}{\sqrt{2}}(1, -1)$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad A = Q \Lambda Q^T$$

18.7 Worked Example 2 (3×3 Diagonalization)

Problem: Diagonalize $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$.

Step 1: Eigenvalues: $\lambda_1 = 1, \lambda_2 = \lambda_3 = 2$ (triangular matrix).

Step 2: Eigenvectors.

For $\lambda_1 = 1$: $(A - I)x = 0 \implies x_1 = (1, 0, 0)$

For $\lambda_2 = 2$: $(A - 2I)x = 0$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} x = 0 \implies x_1 = 0, x_3 = 0, x_2 \text{ free}$$

Only one eigenvector: $x_2 = (0, 1, 0)$.

Conclusion: Only 2 independent eigenvectors for a 3×3 matrix. **Not diagonalizable.**

18.8 Worked Example 3 (Using Diagonalization to Compute A^{10})

Problem: For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, compute A^{10} .

From Example 18.6: $A = S\Lambda S^{-1}$ with $\Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$.

$$\begin{aligned} A^{10} &= S\Lambda^{10}S^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3^{10} & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 3^{10} + 1 & 3^{10} - 1 \\ 3^{10} - 1 & 3^{10} + 1 \end{bmatrix} \end{aligned}$$

Since $3^{10} = 59049$:

$$A^{10} = \frac{1}{2} \begin{bmatrix} 59050 & 59048 \\ 59048 & 59050 \end{bmatrix} = \begin{bmatrix} 29525 & 29524 \\ 29524 & 29525 \end{bmatrix}$$

U3-19. Unit 3 Formula Sheet

Dot Product & Norms

$$x^T y = \sum x_i y_i, \quad \|x\| = \sqrt{x^T x}, \quad \cos \theta = \frac{x^T y}{\|x\| \|y\|}$$

$$x \perp y \iff x^T y = 0, \quad \|x + y\|^2 = \|x\|^2 + \|y\|^2 \text{ (if } x \perp y \text{)}$$

Orthogonal Complement

$$V^\perp = \{x : x^T v = 0 \ \forall v \in V\}, \quad \dim(V) + \dim(V^\perp) = n$$

Fundamental Theorem

$$C(A) \perp N(A^T), \quad C(A^T) \perp N(A)$$

Projections

$$\text{proj}_a(b) = \frac{a^T b}{a^T a} a, \quad P_{\text{line}} = \frac{a a^T}{a^T a}$$

$$P_{C(A)} = A(A^T A)^{-1} A^T, \quad p = P b, \quad q = b - p \in N(A^T)$$

$$P^T = P, \quad P^2 = P \quad (\text{projection matrix properties})$$

Least Squares

$$A^T A \hat{x} = A^T b \quad (\text{normal equations})$$

$$\hat{x} = (A^T A)^{-1} A^T b, \quad e = b - A \hat{x} \perp C(A)$$

$$\text{Weighted: } A^T W A \hat{x} = A^T W b$$

Gram–Schmidt

$$u_1 = v_1, \quad u_k = v_k - \sum_{j=1}^{k-1} \frac{u_j^T v_k}{u_j^T u_j} u_j, \quad q_i = \frac{u_i}{\|u_i\|}$$

QR Factorization

$$A = QR, \quad Q^T Q = I, \quad R \text{ upper triangular}$$

Eigenvalues

$$Ax = \lambda x, \quad \det(A - \lambda I) = 0$$

$$\text{tr}(A) = \sum \lambda_i, \quad \det(A) = \prod \lambda_i$$

Cayley–Hamilton

$$p(A) = 0 \quad \text{where } p(\lambda) = \det(\lambda I - A)$$

Diagonalization

$$A = S \Lambda S^{-1}, \quad A^k = S \Lambda^k S^{-1}$$

$$\text{Symmetric: } A = Q \Lambda Q^T, \quad Q^T Q = I$$

UNIT 4: SINGULAR VALUE DECOMPOSITION, POSITIVE DEFINITE MATRICES & PCA

U4-1. Quadratic Forms

1.1 Definition

A **quadratic form** in n variables is a function:

$$f(x) = x^T A x = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

where A is an $n \times n$ **symmetric** matrix and $x \in \mathbb{R}^n$.

Why symmetric? For any A , $x^T A x = x^T \left(\frac{A+A^T}{2} \right) x$, so we always use the symmetric part.

1.2 Reading the Matrix from a Quadratic Form

For symmetric $A = [a_{ij}]$:

$$x^T A x = \sum_i a_{ii} x_i^2 + 2 \sum_{i < j} a_{ij} x_i x_j$$

Rule:

- Coefficient of $x_i^2 \rightarrow$ diagonal entry a_{ii}
- Coefficient of $x_i x_j$ (for $i \neq j$) $\rightarrow 2a_{ij}$ (split equally: $a_{ij} = a_{ji}$)

1.3 Worked Example 1 (Read matrix from form)

Problem: Find the symmetric matrix for $f = 2x_1^2 + 3x_2^2 - 4x_1x_2$.

- $a_{11} = 2, a_{22} = 3$
- $2a_{12} = -4 \implies a_{12} = -2$

$$A = \begin{bmatrix} 2 & -2 \\ -2 & 3 \end{bmatrix}$$

Verify: $x^T Ax = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2x_1^2 - 4x_1x_2 + 3x_2^2 \checkmark$

1.4 Worked Example 2 (3×3 form)

Problem: Find the symmetric matrix for:

$$f_1 = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$

- $a_{11} = 1, a_{22} = 1, a_{33} = 1$
- $2a_{12} = -2 \implies a_{12} = -1$
- $2a_{13} = -2 \implies a_{13} = -1$
- $2a_{23} = +2 \implies a_{23} = +1$

$$A_1 = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

1.5 Worked Example 3 (Another 3×3)

$$f_2 = x_1^2 + 2x_2^2 + 11x_3^2 - 2x_1x_2 - 2x_1x_3 - 4x_2x_3$$

- $a_{11} = 1, a_{22} = 2, a_{33} = 11$
- $2a_{12} = -2 \implies a_{12} = -1$
- $2a_{13} = -2 \implies a_{13} = -1$

• $2a_{23} = -4 \implies a_{23} = -2$

$$A_2 = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & -2 \\ -1 & -2 & 11 \end{bmatrix}$$

1.6 Completing the Square (Geometric Insight)

For $f = ax_1^2 + 2bx_1x_2 + cx_2^2$:

$$f = a \left(x_1 + \frac{b}{a}x_2\right)^2 + \left(c - \frac{b^2}{a}\right)x_2^2$$

The second coefficient $c - b^2/a$ is the **second pivot** of A . This connects quadratic forms to elimination!

U4-2. Definiteness — Definitions and Geometry

2.1 Definitions (Symmetric Matrix A)

Type	Condition	Meaning
Positive Definite (PD)	$x^T Ax > 0$ for all $x \neq 0$	Always positive
Positive Semidefinite (PSD)	$x^T Ax \geq 0$ for all x	Non-negative (can be zero)
Negative Definite (ND)	$x^T Ax < 0$ for all $x \neq 0$	Always negative
Negative Semidefinite (NSD)	$x^T Ax \leq 0$ for all x	Non-positive
Indefinite	Takes both positive and negative	Neither

Type	Condition	Meaning
	values	

2.2 Geometric Interpretation

- **PD:** The quadratic form $f(x) = x^T Ax$ is a "bowl" (elliptic paraboloid) — minimum at origin
- **ND:** An "inverted bowl" — maximum at origin
- **Indefinite:** A "saddle" — neither min nor max at origin
- **PSD:** A "flat bowl" — touches zero along some direction

PD (bowl):	ND (inverted):	Indefinite (saddle):
<pre> / \ / \ / \ / \ min </pre>	<pre> \ / \ / \ / \ / max </pre>	<pre> / \ / \ / \ / \ saddle </pre>

2.3 Connection to Eigenvalues

For symmetric A with eigenvalues $\lambda_1, \dots, \lambda_n$:

Definiteness	Eigenvalue condition
PD	All $\lambda_i > 0$
PSD	All $\lambda_i \geq 0$
ND	All $\lambda_i < 0$
NSD	All $\lambda_i \leq 0$
Indefinite	Some $\lambda_i > 0$, some $\lambda_i < 0$

2.4 Quick Checks for 2×2 Matrices

For $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$:

$$\text{PD} \iff a > 0 \text{ and } ac - b^2 > 0$$

$$\text{ND} \iff a < 0 \text{ and } ac - b^2 > 0$$

$$\text{Indefinite} \iff ac - b^2 < 0$$

$$\text{PSD} \iff a \geq 0 \text{ and } ac - b^2 = 0 \text{ (and } a, c \geq 0)$$

2.5 Worked Examples (2×2 Checks)

(a) $A = \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$: $a = 1 > 0$, $\det = 5 - 9 = -4 < 0 \rightarrow$ **Indefinite**

(b) $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$: $a = 1 > 0$, $\det = 1 - 1 = 0 \rightarrow$ **PSD (not PD)**

(c) $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$: $a = 2 > 0$, $\det = 10 - 9 = 1 > 0 \rightarrow$ **PD**

(d) $A = \begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix}$: $a = -1 < 0$, $\det = 8 - 4 = 4 > 0 \rightarrow$ **ND**

U4-3. Tests for Positive Definiteness

3.1 Three Equivalent Tests (for Symmetric Matrices)

Test 1 — Eigenvalue Test:

$$A \text{ is PD} \iff \text{all eigenvalues of } A \text{ are } > 0$$

Test 2 — Sylvester's Criterion (Leading Principal Minors):

$$A \text{ is PD} \iff \det(A_1) > 0, \det(A_2) > 0, \dots, \det(A_n) > 0$$

where A_k is the top-left $k \times k$ submatrix.

Test 3 — Pivot Test:

$$A \text{ is PD} \iff \text{all pivots in Gaussian elimination are } > 0$$

All three tests are equivalent for symmetric matrices.

3.2 Worked Example — All Three Tests

Problem: Test $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ for positive definiteness.

Test 1 (Eigenvalues):

$$\det(B - \lambda I) = -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = -(\lambda - 1)^2(\lambda - 4)$$

Eigenvalues: 1, 1, 4 — all positive $\rightarrow$ **PD**

Test 2 (Principal Minors):

$$\det(B_1) = 2 > 0$$

$$\det(B_2) = \det \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = 4 - 1 = 3 > 0$$

$$\det(B_3) = \det(B) = 2(4 - 1) - (-1)(-2 - (-1)) + (-1)(-1 - (-2)) = 6 - 1 - 1 = 4 > 0$$

All positive $\rightarrow$ **PD** ✓

Test 3 (Pivots):

Eliminate: first pivot = 2.

After eliminating column 1:

$$\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & 1/2 \\ 0 & 1/2 & 3/2 \end{bmatrix}$$

Second pivot = $3/2 > 0$.

After eliminating column 2:

$$\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & 1/2 \\ 0 & 0 & 4/3 \end{bmatrix}$$

Third pivot = $4/3 > 0$.

All pivots positive → **PD** ✓

3.3 Worked Example — PSD Matrix (Pivot = 0)

Problem: Test $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ for positive definiteness.

Elimination:

First pivot = 2. After step 1:

$$\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & -3/2 & 3/2 \end{bmatrix}$$

Second pivot = $3/2 > 0$. After step 2:

$$\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & 0 & 0 \end{bmatrix}$$

Third pivot = **0**.

Pivots: 2, $3/2$, 0 — not all positive → **PSD (not PD)**

Note: $\det(A) = 0$ (singular matrix), confirming PSD.

3.4 Theorem: A PD $\implies A^{-1}$ PD

Proof:

Let A be symmetric PD. Take any nonzero x , let $y = A^{-1}x$ (so $y \neq 0$).

$$x^T A^{-1} x = (Ay)^T A^{-1} (Ay) = y^T A^T A^{-1} Ay = y^T Ay > 0$$

(since A is PD and $y \neq 0$). Therefore A^{-1} is PD. $\square$

3.5 Worked Example — Eigenvalue Method for 3×3

Problem: Test $C = M^2$ where $M = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ for positive definiteness.

Key insight: If M has eigenvalues λ , then $C = M^2$ has eigenvalues $\lambda^2 \geq 0$.

Find eigenvalues of M :

$$\det(M - \lambda I) = -\lambda^3 + 6\lambda + 4 = -(\lambda + 2)(\lambda^2 - 2\lambda - 2)$$

Eigenvalues of M : $-2, 1 + \sqrt{3}, 1 - \sqrt{3}$

Eigenvalues of $C = M^2$: $4, (1 + \sqrt{3})^2 = 4 + 2\sqrt{3}, (1 - \sqrt{3})^2 = 4 - 2\sqrt{3}$

All are positive (since $4 - 2\sqrt{3} \approx 0.54 > 0$) $\rightarrow C$ is PD ✓

U4-4. Cholesky Decomposition and LDL^T

4.1 What is Cholesky Decomposition?

If A is symmetric **positive definite**, it can be factored as:

$$A = R^T R$$

where R is **upper triangular** with **positive diagonal entries**.

Equivalently: $A = LL^T$ where L is lower triangular.

Why it exists: PD guarantees all pivots are positive, so we can take their square roots.

4.2 Connection to LDL^T

Gaussian elimination on a symmetric PD matrix gives:

$$A = LDL^T$$

where L is unit lower triangular and $D = \text{diag}(d_1, d_2, \dots, d_n)$ has the pivots.

Then:

$$A = L\sqrt{D}\sqrt{D}L^T = (L\sqrt{D})(L\sqrt{D})^T = R^T R$$

where $R = \sqrt{D}L^T$ is upper triangular.

4.3 Worked Example 1 — $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$

Method 1: Via LDL^T

Elimination:

- First pivot: $d_1 = 5$
- Multiplier: $l_{21} = 4/5$
- Second pivot: $d_2 = 5 - (4/5)^2 \cdot 5 = 5 - 16/5 = 9/5$

$$L = \begin{bmatrix} 1 & 0 \\ \frac{4}{5} & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & \frac{9}{5} \end{bmatrix}$$

$$\sqrt{D} = \begin{bmatrix} \sqrt{5} & 0 \\ 0 & \frac{3}{\sqrt{5}} \end{bmatrix}$$

$$R = \sqrt{D}L^T = \begin{bmatrix} \sqrt{5} & 0 \\ 0 & \frac{3}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} 1 & \frac{4}{5} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{5} & \frac{4}{\sqrt{5}} \\ 0 & \frac{3}{\sqrt{5}} \end{bmatrix}$$

Verify: $R^T R = \begin{bmatrix} \sqrt{5} & 0 \\ \frac{4}{\sqrt{5}} & \frac{3}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \sqrt{5} & \frac{4}{\sqrt{5}} \\ 0 & \frac{3}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & \frac{16}{5} + \frac{9}{5} \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \checkmark$

Method 2: Via Eigendecomposition

For $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$: eigenvalues are $a + b$ and $a - b$.

Eigenvalues: 9 and 1. Eigenvectors: $(1, 1)$ and $(1, -1)$.

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}, \quad \sqrt{\Lambda} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Symmetric square root:

$$A^{1/2} = Q\sqrt{\Lambda}Q^T = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Verify: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \checkmark$

4.4 Worked Example 2 — $A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$

Elimination:

- $d_1 = 4, l_{21} = 2/4 = 1/2$
- $d_2 = 3 - (1/2)^2 \cdot 4 = 3 - 1 = 2$

$$L = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

$$R = \sqrt{D}L^T = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{bmatrix}$$

Cholesky factor: $R = \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{bmatrix}$

4.5 Cholesky Algorithm (General Formula)

For $A = [a_{ij}]$, the Cholesky factor $R = [r_{ij}]$ (upper triangular) is computed as:

$$r_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} r_{ki}^2}$$

$$r_{ij} = \frac{1}{r_{ii}} \left(a_{ij} - \sum_{k=1}^{i-1} r_{ki} r_{kj} \right) \quad (j > i)$$

U4-5. Singular Values — Definition and Meaning

5.1 What are Singular Values?

For any $m \times n$ matrix A (not necessarily square or symmetric):

The **singular values** $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ are defined as:

$$\sigma_i = \sqrt{\lambda_i(A^T A)} = \sqrt{\lambda_i(AA^T)}$$

where λ_i are the **positive eigenvalues** of $A^T A$ (or AA^T).

5.2 Key Facts

- $A^T A$ is always symmetric and **positive semidefinite** (eigenvalues ≥ 0)
- AA^T is also symmetric and PSD
- $A^T A$ and AA^T have the **same nonzero eigenvalues**
- The number of nonzero singular values = rank of A

5.3 Geometric Meaning

Singular values tell you how much A **stretches** space in different directions:

- σ_1 = maximum stretching factor
- σ_r = minimum nonzero stretching factor
- Directions of stretching = singular vectors

5.4 Worked Example

Problem: Find the singular values of $A = \begin{bmatrix} 3 & 0 \\ 4 & 0 \end{bmatrix}$.

$$A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 25 & 0 \\ 0 & 0 \end{bmatrix}$$

Eigenvalues of $A^T A$: 25 and 0.

Singular values: $\sigma_1 = 5$, $\sigma_2 = 0$.

Note: $\sigma_2 = 0$ because A has rank 1.

U4-6. Singular Value Decomposition (SVD)

6.1 The SVD Theorem

Every $m \times n$ matrix A of rank r can be written as:

$$A = U \Sigma V^T$$

where:

- U is $m \times m$ **orthogonal** (columns = left singular vectors)
- Σ is $m \times n$ **diagonal** (singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$)
- V is $n \times n$ **orthogonal** (columns = right singular vectors)

6.2 Structure of Σ

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & & & & 0 & \\ & & & & & \ddots \end{bmatrix}_{m \times n}$$

Only the first r diagonal entries are nonzero.

6.3 The Four Subspaces via SVD

Subspace	Basis
Column space $C(A)$	First r columns of U
Left null space $N(A^T)$	Last $m - r$ columns of U
Row space $C(A^T)$	First r columns of V
Null space $N(A)$	Last $n - r$ columns of V

6.4 Step-by-Step SVD Algorithm

Step 1: Compute $A^T A$ (size $n \times n$).

Step 2: Find eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ of $A^T A$.

Step 3: Singular values: $\sigma_i = \sqrt{\lambda_i}$.

Step 4: Find orthonormal eigenvectors $v_1, v_2, \dots, v_n$ of $A^T A \rightarrow$ these form V .

Step 5: For each nonzero σ_i , compute: $u_i = \frac{Av_i}{\sigma_i}$

Step 6: Extend $\{u_1, \dots, u_r\}$ to a full orthonormal basis of $\mathbb{R}^m \rightarrow$ these form U .

Step 7: Write $A = U\Sigma V^T$.

6.5 Alternative: Use AA^T (when $m < n$)

If $m < n$, it's easier to compute AA^T (size $m \times m$):

- Find eigenvalues and eigenvectors of $AA^T \rightarrow$ these give U
 - Then $v_i = \frac{A^T u_i}{\sigma_i}$ gives V
-
-

U4-7. SVD — Full Worked Examples

Example 7.1 — SVD of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$ (2×3 matrix)

Step 1: Compute AA^T (since $m = 2 < n = 3$, this is easier).

$$AA^T = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 9+4+4 & 6+6-4 \\ 6+6-4 & 4+9+4 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

Step 2: Eigenvalues of AA^T .

For $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$: eigenvalues are $a + b$ and $a - b$.

$$\lambda_1 = 17 + 8 = 25, \lambda_2 = 17 - 8 = 9$$

Singular values: $\sigma_1 = 5, \sigma_2 = 3$

Step 3: Left singular vectors U (eigenvectors of AA^T).

For $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$: eigenvectors are $(1, 1)$ and $(1, -1)$.

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 4: Right singular vectors V (from $v_i = A^T u_i / \sigma_i$).

$$v_1 = \frac{1}{\sigma_1} A^T u_1 = \frac{1}{5} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{5\sqrt{2}} \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$v_2 = \frac{1}{\sigma_2} A^T u_2 = \frac{1}{3} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

For v_3 : find a unit vector orthogonal to v_1 and v_2 .

$$v_3 = \frac{1}{3} \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} \text{ (verify: } v_1^T v_3 = 0, v_2^T v_3 = 0, \|v_3\| = 1 \text{)}$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ 0 & \frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{bmatrix}$$

Step 5: Σ (2×3 matrix).

$$\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}$$

Final SVD: $A = U \Sigma V^T$

Example 7.2 — SVD of $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}$ (3×2 matrix)

Step 1: Compute $A^T A$ (since $n = 2 < m = 3$, this is easier).

Columns: $c_1 = (1, -2, 2)$, $c_2 = (-1, 2, 2)$.

$$A^T A = \begin{bmatrix} c_1 \cdot c_1 & c_1 \cdot c_2 \\ c_2 \cdot c_1 & c_2 \cdot c_2 \end{bmatrix} = \begin{bmatrix} 1+4+4 & -1-4+4 \\ -1-4+4 & 1+4+4 \end{bmatrix} = \begin{bmatrix} 9 & -1 \\ -1 & 9 \end{bmatrix}$$

Step 2: Eigenvalues of $A^T A$.

$$\lambda_1 = 9 + 1 = 10, \lambda_2 = 9 - 1 = 8$$

Singular values: $\sigma_1 = \sqrt{10}$, $\sigma_2 = 2\sqrt{2}$

Step 3: Right singular vectors V .

For eigenvalue 10: $(A^T A - 10I)v = 0 \implies -v_1 - v_2 = 0 \implies v_1 = -v_2$

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

For eigenvalue 8: $(A^T A - 8I)v = 0 \implies v_1 - v_2 = 0 \implies v_1 = v_2$

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

Step 4: Left singular vectors U .

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{20}} \begin{bmatrix} 2 \\ -4 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$$

$$u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

For u_3 : orthogonal to u_1 and u_2 : $u_3 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

$$U = \begin{bmatrix} \frac{1}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \\ 0 & 1 & 0 \end{bmatrix}$$

Step 5: Σ (3×2 matrix).

$$\Sigma = \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 2\sqrt{2} \\ 0 & 0 \end{bmatrix}$$

Example 7.3 — SVD of Symmetric PD Matrix $A = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$

For a **symmetric PD** matrix, the SVD is the same as the eigendecomposition:

$$A = Q\Lambda Q^T = U\Sigma V^T \quad \text{with } U = V = Q, \Sigma = \Lambda$$

Eigenvalues:

$$\det(A - \lambda I) = (3 - \lambda)(2 - \lambda) - 1 = \lambda^2 - 5\lambda + 5 = 0$$

$$\lambda_{1,2} = \frac{5 \pm \sqrt{5}}{2}$$

Both positive (PD confirmed). These are also the singular values.

Eigenvectors: Solve $(A - \lambda_i I)x = 0$ for each λ_i .

$$\text{For } \lambda_1 = \frac{5+\sqrt{5}}{2}: (3 - \lambda_1)x_1 = x_2 \implies x_1 = \begin{bmatrix} 1 \\ \lambda_1 - 3 \end{bmatrix} = \begin{bmatrix} 2 \\ \sqrt{5} - 1 \end{bmatrix} \text{ (up to scaling)}$$

Normalize to get q_1 , and similarly q_2 .

SVD: $A = Q\Lambda Q^T$ with $Q = [q_1 | q_2]$ and $\Lambda = \text{diag}(\lambda_1, \lambda_2)$.

Example 7.4 — SVD of a Rank-1 Matrix

Problem: Find SVD of $A = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix}$.

Since A is rank 1, there is only one nonzero singular value.

$$A^T A = \begin{bmatrix} 6 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 45 & 15 \\ 15 & 5 \end{bmatrix}$$

Eigenvalues: $\lambda_1 = 50$, $\lambda_2 = 0$ (trace = 50, det = 0).

$$\sigma_1 = \sqrt{50} = 5\sqrt{2}$$

$$v_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \text{ (eigenvector for } \lambda_1 = 50 \text{)}$$

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{5\sqrt{2}} \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \frac{1}{5\sqrt{20}} \begin{bmatrix} 20 \\ 10 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A = u_1 \sigma_1 v_1^T = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot 5\sqrt{2} \cdot \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} \checkmark$$

U4-8. Low-Rank Approximation (Image Compression Idea)

8.1 The SVD Expansion

The SVD gives a sum of rank-1 matrices:

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T$$

Each term $\sigma_i u_i v_i^T$ is a rank-1 matrix. The terms are ordered by importance ($\sigma_1 \geq \sigma_2 \geq \dots$).

8.2 Best Rank- k Approximation

The **best rank- k approximation** to A (in the sense of minimizing $\|A - A_k\|$) is:

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$$

This is the **Eckart–Young theorem**.

8.3 Why This Matters for Image Compression

An image is a matrix of pixel values. If the singular values decay quickly:

- Keep only the first k terms
- Storage: $k(m + n + 1)$ numbers instead of mn
- Compression ratio: $\frac{mn}{k(m+n+1)}$

8.4 Worked Example

Problem: For $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$ (from Example 7.1), write the rank-1 approximation.

From Example 7.1: $\sigma_1 = 5$, $u_1 = \frac{1}{\sqrt{2}}(1, 1)^T$, $v_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T$.

$$A_1 = \sigma_1 u_1 v_1^T = 5 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} = \frac{5}{2} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2.5 & 2.5 & 0 \\ 2.5 & 2.5 & 0 \end{bmatrix}$$

Error: $\|A - A_1\| = \sigma_2 = 3$ (the next singular value).

U4-9. Principal Component Analysis (PCA) — Theory

9.1 What is PCA?

PCA is a **dimensionality reduction** technique that finds the directions of **maximum variance** in data.

Goal: Given high-dimensional data, find a lower-dimensional representation that preserves as much information (variance) as possible.

9.2 Key Concepts

- **Principal Components (PCs):** Orthogonal directions of maximum variance
- **PC1:** Direction of maximum variance
- **PC2:** Direction of maximum variance, orthogonal to PC1
- **Scores:** Projections of data onto principal components
- **Explained variance:** How much of total variance each PC captures

9.3 Two Methods

Method 1: PCA via SVD

1. Center the data: $\tilde{X} = X - \text{mean}$
2. Compute SVD: $\tilde{X} = U\Sigma V^T$
3. Principal directions = columns of V
4. Scores = $\tilde{X}V = U\Sigma$

Method 2: PCA via Covariance Matrix

1. Center the data: $\tilde{X} = X - \text{mean}$
2. Compute covariance matrix C
3. Find eigenvectors of C
4. Principal directions = eigenvectors (sorted by eigenvalue)

9.4 Data Centering Rules (from your course notes)

Matrix shape	Rule
Rows = features, Columns = samples (rows < columns)	Subtract row means
Rows = samples, Columns = features (rows > columns)	Subtract column means

General rule: Subtract the mean of whichever dimension contains the **features**.

9.5 Covariance Matrix Formula

If rows are samples and columns are features (n samples, p features):

$$C = \frac{1}{n-1} \tilde{X}^T \tilde{X} \quad (\text{size } p \times p)$$

If rows are features and columns are samples (p features, n samples):

$$C = \frac{1}{n-1} \tilde{X} \tilde{X}^T \quad (\text{size } p \times p)$$

9.6 Explained Variance

If eigenvalues of C are $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$:

$$\text{Variance explained by PC}_k = \frac{\lambda_k}{\sum_{i=1}^p \lambda_i} \times 100\%$$

U4-10. PCA via SVD — Worked Examples

Example 10.1 — PCA via SVD (Simple 2×2 Data)

Problem: Given $X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ (rows = samples, columns = features), perform PCA via SVD.

Step 1: Center the data (subtract column means).

Column means: $\mu_1 = (2 + 0)/2 = 1$, $\mu_2 = (0 + 2)/2 = 1$.

$$\tilde{X} = \begin{bmatrix} 2-1 & 0-1 \\ 0-1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Step 2: SVD of $\tilde{X}$.

$$\tilde{X}^T \tilde{X} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}^T \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

Eigenvalues: $\lambda_1 = 4$, $\lambda_2 = 0$.

Singular values: $\sigma_1 = 2$, $\sigma_2 = 0$.

Eigenvector for $\lambda_1 = 4$: $v_1 = \frac{1}{\sqrt{2}}(1, -1)^T$

Step 3: First principal component.

$$\text{PC1 direction: } w_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Interpretation: The data varies along the direction $(1, -1)$. The second PC has zero variance (data is 1D after centering).

Example 10.2 — PCA via SVD (Larger Dataset)

Problem: Given the movie preference matrix $A \in \mathbb{R}^{5 \times 20}$ (5 features, 20 users), perform PCA.

Since rows (features) < columns (users), we do **row mean centering**.

Step 1: Compute row means μ_i for each feature i .

Step 2: $\tilde{A} = A - \mu$ (subtract row mean from each row).

Step 3: Compute $\tilde{A}\tilde{A}^T$ (5×5 matrix).

Step 4: Find eigenvalues and eigenvectors of $\tilde{A}\tilde{A}^T$.

Step 5: Eigenvectors = principal components (directions in feature space).

Step 6: Project data: scores = $\tilde{A}^T \cdot$ eigenvectors.

U4-11. PCA via Covariance Matrix — Worked Examples

Example 11.1 — Covariance Matrix (2 Features, 6 Students)

Problem: Math and history scores for 6 students:

$$A = \begin{bmatrix} -10 & -5 & 0 & 5 & 10 & 0 \\ -10 & 0 & 10 & -10 & 0 & 10 \end{bmatrix}$$

(Rows = features, Columns = students; already centered since row means = 0)

Step 1: Row means are both 0, so $\tilde{X} = A$.

Step 2: Covariance matrix (rows < columns, so $C = \frac{1}{n-1} \tilde{X} \tilde{X}^T$, $n = 6$):

$$\tilde{X} \tilde{X}^T = \begin{bmatrix} 100 + 25 + 0 + 25 + 100 + 0 & 100 + 0 + 0 - 50 + 0 + 0 \\ 100 + 0 + 0 - 50 + 0 + 0 & 100 + 0 + 100 + 100 + 0 + 100 \end{bmatrix} = \begin{bmatrix} 250 & 50 \\ 50 & 400 \end{bmatrix}$$

$$C = \frac{1}{5} \begin{bmatrix} 250 & 50 \\ 50 & 400 \end{bmatrix} = \begin{bmatrix} 50 & 10 \\ 10 & 80 \end{bmatrix}$$

Step 3: Eigenvalues of C .

$$\det(C - \lambda I) = (50 - \lambda)(80 - \lambda) - 100 = \lambda^2 - 130\lambda + 3900 = 0$$

$$\lambda = \frac{130 \pm \sqrt{16900 - 15600}}{2} = \frac{130 \pm \sqrt{1300}}{2} = 65 \pm \frac{\sqrt{1300}}{2}$$

$$\lambda_1 \approx 83.0, \lambda_2 \approx 47.0$$

Step 4: Explained variance.

$$\text{Total variance} = \text{tr}(C) = 130.$$

PC1 explains $\approx 83/130 \approx 63.8\%$, PC2 explains $\approx 36.2\%$.

Example 11.2 — Full PCA (3 Subjects, 5 Students)

Problem: Student marks:

$$A = \begin{bmatrix} 80 & 85 & 90 \\ 70 & 75 & 78 \\ 60 & 65 & 70 \\ 88 & 82 & 84 \\ 72 & 74 & 76 \end{bmatrix}$$

(Rows = students = samples, Columns = subjects = features; rows > columns $\rightarrow$ column mean centering)

Step 1: Column means.

$$\mu_1 = (80 + 70 + 60 + 88 + 72)/5 = 74$$

$$\mu_2 = (85 + 75 + 65 + 82 + 74)/5 = 76.2$$

$$\mu_3 = (90 + 78 + 70 + 84 + 76)/5 = 79.6$$

Step 2: Center the data.

$$\tilde{X} = \begin{bmatrix} 6 & 8.8 & 10.4 \\ -4 & -1.2 & -1.6 \\ -14 & -11.2 & -9.6 \\ 14 & 5.8 & 4.4 \\ -2 & -2.2 & -3.6 \end{bmatrix}$$

Step 3: Covariance matrix ($C = \frac{1}{n-1} \tilde{X}^T \tilde{X}$, $n = 5$):

$$\tilde{X}^T \tilde{X} = \begin{bmatrix} 448 & 300 & 272 \\ 300 & 242.8 & 234.6 \\ 272 & 234.6 & 235.2 \end{bmatrix}$$

$$C = \frac{1}{4} \tilde{X}^T \tilde{X} = \begin{bmatrix} 112 & 75 & 68 \\ 75 & 60.7 & 58.65 \\ 68 & 58.65 & 58.8 \end{bmatrix}$$

Step 4: Eigenvalues of C .

$$\text{tr}(C) = 112 + 60.7 + 58.8 = 231.5$$

The largest eigenvalue captures most of the variance (the three subjects are highly correlated).

Interpretation: Since all three subjects have similar patterns (students who score high in one tend to score high in others), PC1 will capture most variance and represent "overall academic performance."

Example 11.3 — Complete PCA (2 Features, 3 Samples)

Problem: Dataset $X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 3 & 1 \end{bmatrix}$ (rows = samples, columns = features).

Step 1: Column means.

$$\mu = \left(\frac{2+0+3}{3}, \frac{0+2+1}{3} \right) = \left(\frac{5}{3}, 1 \right)$$

Step 2: Center.

$$\tilde{X} = \begin{bmatrix} 2 - 5/3 & 0 - 1 \\ 0 - 5/3 & 2 - 1 \\ 3 - 5/3 & 1 - 1 \end{bmatrix} = \begin{bmatrix} 1/3 & -1 \\ -5/3 & 1 \\ 4/3 & 0 \end{bmatrix}$$

Step 3: Covariance matrix ($n = 3, n - 1 = 2$):

$$\begin{aligned} \tilde{X}^T \tilde{X} &= \begin{bmatrix} (1/3)^2 + (5/3)^2 + (4/3)^2 & (1/3)(-1) + (-5/3)(1) + (4/3)(0) \\ (1/3)(-1) + (-5/3)(1) + (4/3)(0) & 1 + 1 + 0 \end{bmatrix} \\ &= \begin{bmatrix} 1/9 + 25/9 + 16/9 & -1/3 - 5/3 \\ -1/3 - 5/3 & 2 \end{bmatrix} = \begin{bmatrix} 42/9 & -2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 14/3 & -2 \\ -2 & 2 \end{bmatrix} \\ C &= \frac{1}{2} \begin{bmatrix} 14/3 & -2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 7/3 & -1 \\ -1 & 1 \end{bmatrix} \end{aligned}$$

Step 4: Eigenvalues.

$$\det(C - \lambda I) = \left(\frac{7}{3} - \lambda\right)(1 - \lambda) - 1 = \lambda^2 - \frac{10}{3}\lambda + \frac{4}{3} = 0$$

$$3\lambda^2 - 10\lambda + 4 = 0 \implies \lambda = \frac{10 \pm \sqrt{100 - 48}}{6} = \frac{10 \pm \sqrt{52}}{6} = \frac{5 \pm \sqrt{13}}{3}$$

$$\lambda_1 = \frac{5 + \sqrt{13}}{3} \approx 2.869, \quad \lambda_2 = \frac{5 - \sqrt{13}}{3} \approx 0.464$$

Step 5: Explained variance.

$$\text{Total} = \lambda_1 + \lambda_2 = 10/3.$$

$$\text{PC1: } \frac{5 + \sqrt{13}}{10} \approx 86.1\%, \quad \text{PC2: } \frac{5 - \sqrt{13}}{10} \approx 13.9\%$$

Step 6: Eigenvectors (principal directions).

$$\text{For } \lambda_1: (7/3 - \lambda_1)v_1 = v_2$$

$$v_1 = \frac{1}{\|(3, 2 - \sqrt{13})\|} (3, 2 - \sqrt{13}) \approx (0.882, -0.472)$$

$$\text{For } \lambda_2: v_2 \approx (0.472, 0.882)$$

Step 7: Scores (projections onto PC1).

$$z_1 = \tilde{X}v_1 \approx \begin{bmatrix} 1/3 \\ -5/3 \\ 4/3 \end{bmatrix} (0.882) + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} (-0.472) \approx \begin{bmatrix} 0.766 \\ -1.942 \\ 1.176 \end{bmatrix}$$

U4-12. Minimum Principles and Rayleigh Quotient

12.1 Rayleigh Quotient

For a symmetric matrix A and nonzero vector x :

$$R(x) = \frac{x^T A x}{x^T x}$$

12.2 Minimum and Maximum Principles

$$\lambda_{\min} = \min_{x \neq 0} R(x) = \min_{\|x\|=1} x^T A x$$

$$\lambda_{\max} = \max_{x \neq 0} R(x) = \max_{\|x\|=1} x^T A x$$

The minimum is achieved when $x = q_n$ (eigenvector for $\lambda_{\min}$).

The maximum is achieved when $x = q_1$ (eigenvector for $\lambda_{\max}$).

12.3 Connection to Positive Definiteness

$$A \text{ is PD} \iff \lambda_{\min} > 0 \iff R(x) > 0 \text{ for all } x \neq 0$$

12.4 PCA as Maximum Variance

PCA finds the direction w (unit vector) that maximizes the projected variance:

$$\max_{\|w\|=1} w^T C w = \lambda_{\max}(C)$$

achieved at $w = q_1$ (eigenvector of C for largest eigenvalue).

This is exactly the Rayleigh quotient maximization!

12.5 Worked Example

Problem: For $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, find the minimum and maximum of $R(x)$.

Eigenvalues: $\lambda_1 = 4, \lambda_2 = 2$.

$$\lambda_{\min} = 2 = \min_{\|x\|=1} x^T A x, \quad \lambda_{\max} = 4 = \max_{\|x\|=1} x^T A x$$

Verify at $x = \frac{1}{\sqrt{2}}(1, -1)$ (eigenvector for $\lambda_2 = 2$):

$$x^T Ax = \frac{1}{2}(1, -1) \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2}(1, -1) \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \frac{1}{2}(4) = 2 = \lambda_{\min} \checkmark$$

U4-13. Unit 4 Formula Sheet

Quadratic Forms

$$f(x) = x^T Ax, \quad A \text{ symmetric}$$

$$\text{Coefficient of } x_i^2 = a_{ii}, \quad \text{Coefficient of } x_i x_j = 2a_{ij}$$

Definiteness Tests (Symmetric A)

$$\text{PD} \iff \text{all eigenvalues} > 0 \iff \text{all leading minors} > 0 \iff \text{all pivots} > 0$$

$$\text{PD} \implies A^{-1} \text{ is PD}$$

Cholesky

$$A = R^T R \quad (R \text{ upper triangular, positive diagonal})$$

$$A = LDL^T \quad (L \text{ unit lower triangular, } D \text{ diagonal pivots})$$

Singular Values

$$\sigma_i = \sqrt{\lambda_i(A^T A)}, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0$$

SVD

$$A = U \Sigma V^T, \quad U^T U = I_m, \quad V^T V = I_n$$

$$u_i = \frac{Av_i}{\sigma_i}, \quad v_i = \frac{A^T u_i}{\sigma_i}$$

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T \quad (\text{outer product expansion})$$

Low-Rank Approximation

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^T \quad (\text{best rank-}k \text{ approximation})$$

PCA

$$\tilde{X} = X - \text{mean}, \quad C = \frac{1}{n-1} \tilde{X}^T \tilde{X} \quad (\text{or } \tilde{X} \tilde{X}^T)$$

PCs = eigenvectors of C (sorted by eigenvalue)

$$\text{Explained variance of PC}_k = \frac{\lambda_k}{\sum \lambda_i}$$

Rayleigh Quotient

$$R(x) = \frac{x^T A x}{x^T x}, \quad \lambda_{\min} \leq R(x) \leq \lambda_{\max}$$

PRACTICE SECTION

PRACTICE-1. MCQ Bank (Units 3 & 4)

Instructions: Attempt all questions before checking the answer key. Answers are in PRACTICE-3.

UNIT 3 MCQs (1 mark each)

Q1. If $x \perp y$, then $x^T y$ equals:

- (A) $\|x\|\|y\|$ (B) 1 (C) 0 (D) -1

Q2. $\|x\|^2$ equals:

- (A) $x^T x$ (B) $x^T y$ (C) $x + y$ (D) $\det(x)$

Q3. The projection of b onto the line spanned by a is:

- (A) $\frac{a^T a}{a^T b} a$ (B) $\frac{a^T b}{a^T a} a$ (C) $\frac{b^T b}{a^T a} a$ (D) $\frac{a^T b}{b^T b} b$

Q4. For an orthogonal projection matrix P , which is true?

- (A) $P^T = -P$ (B) $P^2 = I$ (C) $P^2 = P$ (D) $\det(P) = 1$ always

Q5. The normal equations for least squares are:

- (A) $Ax = b$ (B) $AA^T x = A^T b$ (C) $A^T Ax = A^T b$ (D) $A^T Ax = b$

Q6. If columns of Q are orthonormal, then:

- (A) $QQ^T = I$ always (B) $Q^T Q = I$ (C) $Q^{-1} = Q$ (D) $\det(Q) = 0$

Q7. If A is $m \times n$, then $C(A) \subseteq$:

- (A) $\mathbb{R}^n$ (B) $\mathbb{R}^m$ (C) $\mathbb{R}^{m \times n}$ (D) $N(A)$

Q8. $N(A^T)$ is also called:

- (A) Column space (B) Row space (C) Left null space (D) Right null space

Q9. If A is triangular, its eigenvalues are:

- (A) Row sums (B) Column sums (C) Diagonal entries (D) Always 1

Q10. If $Ax = \lambda x$ with $x \neq 0$, then x is:

- (A) A basis vector (B) An eigenvector (C) A pivot vector (D) A null vector

Q11. For symmetric A , eigenvalues are always:

- (A) Complex (B) Real (C) Zero (D) Negative

Q12. For P an orthogonal projection matrix, eigenvalues can be:

- (A) Only 0 or 1 (B) Any real number (C) Only -1 or 1 (D) Only 2

Q13. If Q is orthogonal (square), then Q^{-1} equals:

- (A) Q (B) Q^T (C) $-Q$ (D) $I - Q$

Q14. If vectors u, v are orthogonal, then $\|u + v\|^2$ equals:

- (A) $\|u\|^2 + \|v\|^2$ (B) $\|u\|^2 - \|v\|^2$ (C) $\|u\|\|v\|$ (D) 0

Q15. The characteristic equation comes from:

- (A) $\det(A) = 0$ (B) $\det(A - \lambda I) = 0$ (C) $\|A\| = 0$ (D) $A^T = A$

Q16. A is diagonalizable if it has:

- (A) n linearly independent eigenvectors (B) Only one eigenvalue
(C) $\det(A) = 0$ (D) A is always diagonalizable

Q17. The Cayley–Hamilton theorem states:

- (A) $A^T = A^{-1}$ (B) $p(A) = 0$ where p is the characteristic polynomial
(C) $\det(A) = 0$ (D) $A = Q\Lambda Q^T$

Q18. If A has eigenvalues 2 and 5, then $\det(A)$ equals:

- (A) 7 (B) 10 (C) 3 (D) 25

Q19. The dimension of $V^\perp$ when V is a 2D subspace of $\mathbb{R}^5$ is:

- (A) 2 (B) 3 (C) 5 (D) 1

Q20. In least squares, the residual $e = b - A\hat{x}$ is:

- (A) Parallel to $C(A)$ (B) Orthogonal to $C(A)$ (C) Always zero (D) An eigenvector of A
-

UNIT 3 MCQs (2 marks each)

Q21. Let $a = (1, 1, 0)$ and $b = (2, 0, 0)$. The projection p of b onto $\text{span}(a)$ is:

- (A) $(1, 1, 0)$ (B) $(2, 0, 0)$ (C) $(1, 0, 0)$ (D) $(2, 2, 0)$

Q22. For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, eigenvalues are:

- (A) 2, 2 (B) 3, 1 (C) 4, 0 (D) 1, 1

Q23. If P projects onto a 2D subspace of $\mathbb{R}^3$, the rank of P is:

- (A) 1 (B) 2 (C) 3 (D) 0

Q24. For $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, the trace and determinant are:

- (A) trace = 4, det = 3 (B) trace = 3, det = 4 (C) trace = 4, det = 4 (D) trace = 3, det = 3

Q25. The Gram–Schmidt process converts independent vectors into:

- (A) Eigenvectors (B) Orthonormal vectors (C) Null vectors (D) Row vectors
-

UNIT 4 MCQs (1 mark each)

Q26. A symmetric matrix A is positive definite if:

- (A) $\det(A) > 0$ (B) All eigenvalues > 0 (C) $A^T = A$ (D) $\text{tr}(A) > 0$

Q27. The SVD of A is written as:

- (A) $A = U\Sigma V$ (B) $A = U\Sigma V^T$ (C) $A = V\Sigma U^T$ (D) $A = \Sigma UV^T$

Q28. Singular values of A are:

- (A) Eigenvalues of A (B) Square roots of eigenvalues of $A^T A$
(C) Diagonal entries of A (D) Eigenvalues of A^T

Q29. For a PD matrix A , the Cholesky factorization is:

- (A) $A = LU$ (B) $A = QR$ (C) $A = R^T R$ (D) $A = LDL^T$ only

Q30. In PCA, the first principal component is:

- (A) The mean of the data (B) The direction of maximum variance
(C) The direction of minimum variance (D) The largest eigenvalue

Q31. The quadratic form $x^T A x$ for $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $x = (1, 1)$ equals:

- (A) 5 (B) 7 (C) 6 (D) 8

Q32. If A is PD, then A^{-1} is:

- (A) Negative definite (B) Positive definite (C) Indefinite (D) Singular

Q33. The number of nonzero singular values of A equals:

- (A) $\min(m, n)$ (B) $\max(m, n)$ (C) $\text{rank}(A)$ (D) $m + n$

Q34. In PCA, data must be _____ before computing the covariance matrix:

- (A) Normalized (B) Centered (mean subtracted) (C) Squared (D) Transposed

Q35. The Rayleigh quotient $R(x) = x^T A x / x^T x$ for symmetric A satisfies:

- (A) $R(x) = \det(A)$ (B) $\lambda_{\min} \leq R(x) \leq \lambda_{\max}$
(C) $R(x) = \text{tr}(A)$ (D) $R(x) = 0$ always
-

UNIT 4 MCQs (2 marks each)

Q36. For $A = \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$, which is true?

- (A) PD (B) PSD (C) Indefinite (D) ND

Q37. The singular values of $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ are:

- (A) 3, 4 (B) 9, 16 (C) 5, 0 (D) 7, 0

Q38. For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, the Cholesky factor R (upper triangular) has r_{11} equal to:

- (A) $\sqrt{2}$ (B) 2 (C) 1 (D) $\sqrt{3}$

Q39. If the covariance matrix has eigenvalues 8 and 2, the variance explained by PC1 is:

- (A) 80% (B) 75% (C) 60% (D) 50%

Q40. The best rank-1 approximation to A uses:

- (A) The smallest singular value (B) The largest singular value
(C) All singular values (D) The mean of singular values

PRACTICE-2. Numerical Problem Bank

Instructions: Solve all problems completely before checking solutions. Full solutions are in PRACTICE-3.

UNIT 3 — Numerical Problems

N1. (Orthogonality — Basic)

- (a) Show that $v_1 = (1, 2, -1)$ and $v_2 = (2, -1, 0)$ are orthogonal.
(b) Find a vector in $\mathbb{R}^3$ orthogonal to both $(1, 1, 0)$ and $(1, 0, 1)$.

N2. (Orthogonal Complement)

Find the orthogonal complement of $V = \text{span}\{(1, 2, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

N3. (Projection onto a Line)

Find the projection of $b = (3, 4, 0)$ onto the line spanned by $a = (1, 1, 1)$.

N4. (Projection onto Column Space)

Find the projection of $b = (1, 2, 7)$ onto the column space of $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}$.

Split $b = p + q$ where $p \in C(A)$ and $q \in N(A^T)$.

N5. (Least Squares Line Fit)

Fit the best line $y = C + Dt$ to the data points $(0, 1), (1, 2), (2, 2), (3, 3)$.

N6. (Least Squares — Overdetermined System)

Find the least squares solution to:

$$x_1 + x_2 = 2, \quad x_1 - x_2 = 0, \quad x_1 + 2x_2 = 3$$

N7. (Gram–Schmidt in $\mathbb{R}^3$)

Apply Gram–Schmidt to $v_1 = (1, 0, 1), v_2 = (1, 1, 0), v_3 = (0, 1, 1)$.

Find both the orthogonal and orthonormal bases.

N8. (Gram–Schmidt in $\mathbb{R}^4$)

Apply Gram–Schmidt to $v_1 = (1, 1, 1, 1)$, $v_2 = (1, 2, 4, 5)$, $v_3 = (1, -3, -4, -2)$.

N9. (Eigenvalues and Eigenvectors)

Find all eigenvalues and eigenvectors of:

(a) $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

(b) $B = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

(c) $C = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 2 & 4 \end{bmatrix}$

N10. (Cayley–Hamilton)

(a) For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, verify Cayley–Hamilton and use it to find A^{-1} .

(b) For $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, use Cayley–Hamilton to find A^3 .

N11. (Diagonalization)

Diagonalize $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. Write $A = S\Lambda S^{-1}$ and compute A^5 .

N12. (Orthogonal Diagonalization)

Orthogonally diagonalize $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$. Write $A = Q\Lambda Q^T$.

N13. (Weighted Least Squares)

Fit $y = C + Dt$ to $(0, 1)$, $(1, 3)$, $(2, 4)$ with weights $w = (2, 1, 2)$.

UNIT 4 — Numerical Problems

N14. (Quadratic Forms)

Write the symmetric matrix for each quadratic form:

(a) $f = 3x_1^2 + 2x_2^2 - 4x_1x_2$

(b) $g = x_1^2 + 4x_2^2 + x_3^2 + 2x_1x_2 - 6x_1x_3 + 4x_2x_3$

N15. (Definiteness Tests)

Test each matrix for positive definiteness using all three methods (eigenvalues, minors, pivots):

$$(a) A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$(b) B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

$$(c) C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

N16. (Cholesky Decomposition)

Find the Cholesky factorization $A = R^T R$ for:

$$(a) A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}$$

N17. (SVD — Full)

Find the complete SVD (U , Σ , V) for:

$$(a) A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$(c) A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

N18. (PCA via Covariance)

Given the dataset (rows = students, columns = subjects):

$$X = \begin{bmatrix} 80 & 85 \\ 70 & 75 \\ 60 & 65 \\ 88 & 82 \\ 72 & 74 \end{bmatrix}$$

(a) Center the data

(b) Compute the covariance matrix

- (c) Find eigenvalues and principal directions
- (d) Compute the explained variance ratio

N19. (PCA via SVD)

Given $X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ (rows = samples, columns = features):

- (a) Center the data
- (b) Find the SVD of the centered matrix
- (c) Identify the first principal component

N20. (Rayleigh Quotient)

For $A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$:

- (a) Find the eigenvalues
- (b) Find the maximum and minimum of $R(x) = x^T A x / x^T x$
- (c) At which unit vectors are these achieved?

PRACTICE-3. Answer Key + Full Solutions

MCQ Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans
Q1	C	Q11	B	Q21	A	Q31	B
Q2	A	Q12	A	Q22	B	Q32	B
Q3	B	Q13	B	Q23	B	Q33	C
Q4	C	Q14	A	Q24	A	Q34	B
Q5	C	Q15	B	Q25	B	Q35	B

Q	Ans	Q	Ans	Q	Ans	Q	Ans
Q6	B	Q16	A	Q26	B	Q36	C
Q7	B	Q17	B	Q27	B	Q37	A
Q8	C	Q18	B	Q28	B	Q38	A
Q9	C	Q19	B	Q29	C	Q39	A
Q10	B	Q20	B	Q30	B	Q40	B

MCQ Explanations for Tricky Questions

Q21: $a^T b = 2$, $a^T a = 2$, so $p = (2/2)(1, 1, 0) = (1, 1, 0)$. Answer: A.

Q24: Trace = $1 + 3 = 4$, $\det = 3 - 0 = 3$. Answer: A.

Q31: $x^T A x = (1, 1) \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1, 1) \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 7$. Answer: B.

Q36: $a = 1 > 0$, $\det = 5 - 9 = -4 < 0 \rightarrow$ Indefinite. Answer: C.

Q37: A is diagonal, so singular values = $|\text{diagonal entries}| = 3, 4$. Answer: A.

Q38: First pivot = $a_{11} = 2$, so $r_{11} = \sqrt{2}$. Answer: A.

Q39: PC1 variance = $8/(8 + 2) = 80\%$. Answer: A.

Full Solutions to Numerical Problems

N1 Solution:

(a) $v_1^T v_2 = (1)(2) + (2)(-1) + (-1)(0) = 2 - 2 + 0 = 0 \checkmark$ Orthogonal.

(b) Let $x = (x_1, x_2, x_3)$.

$$x_1 + x_2 = 0 \text{ and } x_1 + x_3 = 0.$$

$$\text{Set } x_1 = 1: x = (1, -1, -1).$$

N2 Solution:

$$\text{Let } x = (x_1, x_2, x_3).$$

$$x \cdot (1, 2, 0) = 0 \implies x_1 + 2x_2 = 0$$

$$x \cdot (0, 1, 1) = 0 \implies x_2 + x_3 = 0$$

$$\text{From first: } x_1 = -2x_2. \text{ From second: } x_3 = -x_2.$$

$$V^\perp = \text{span}\{(-2, 1, -1)\}$$

$$\text{Check: } \dim(V) = 2, \dim(V^\perp) = 1, 2 + 1 = 3 \checkmark$$

N3 Solution:

$$a = (1, 1, 1), b = (3, 4, 0).$$

$$a^T b = 3 + 4 + 0 = 7, a^T a = 3.$$

$$p = \frac{7}{3}(1, 1, 1) = \left(\frac{7}{3}, \frac{7}{3}, \frac{7}{3}\right)$$

$$q = b - p = \left(\frac{2}{3}, \frac{5}{3}, -\frac{7}{3}\right)$$

$$\text{Verify: } a^T q = \frac{2}{3} + \frac{5}{3} - \frac{7}{3} = 0 \checkmark$$

N4 Solution:

This is the full worked example from U3-7.5.

$$\hat{x} = (9/22, 37/22), p = (23/11, -14/11, 65/11), q = (-12/11, 36/11, 12/11).$$

N5 Solution:

This is the full worked example from U3-10.2.

Best fit line: $y = 1.1 + 0.6t$.

N6 Solution:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}, A^T b = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

(Check: $A^T b$ row 1 = $1(2) + 1(0) + 1(3) = 5$; row 2 = $1(2) + (-1)(0) + 2(3) = 8$)

$$\det = 18 - 4 = 14$$

$$\hat{x} = \frac{1}{14} \begin{bmatrix} 6 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 14 \\ 14 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

N7 Solution:

$$v_1 = (1, 0, 1), v_2 = (1, 1, 0), v_3 = (0, 1, 1).$$

Step 1: $u_1 = (1, 0, 1), \|u_1\|^2 = 2$

Step 2: $v_2^T u_1 = 1, \text{proj}_{u_1}(v_2) = \frac{1}{2}(1, 0, 1)$

$$u_2 = (1, 1, 0) - \frac{1}{2}(1, 0, 1) = \left(\frac{1}{2}, 1, -\frac{1}{2}\right), \|u_2\|^2 = \frac{3}{2}$$

Step 3: $v_3^T u_1 = 1, \text{proj}_{u_1}(v_3) = \frac{1}{2}(1, 0, 1)$

$$v_3^T u_2 = 0 + 1 - \frac{1}{2} = \frac{1}{2}, \text{proj}_{u_2}(v_3) = \frac{1/2}{3/2} u_2 = \frac{1}{3} \left(\frac{1}{2}, 1, -\frac{1}{2}\right)$$

$$u_3 = (0, 1, 1) - \frac{1}{2}(1, 0, 1) - \frac{1}{3} \left(\frac{1}{2}, 1, -\frac{1}{2}\right) = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

Orthogonal basis: $\{(1, 0, 1), (\frac{1}{2}, 1, -\frac{1}{2}), (-\frac{2}{3}, \frac{2}{3}, \frac{2}{3})\}$

Orthonormal basis:

$$q_1 = \frac{1}{\sqrt{2}}(1, 0, 1), q_2 = \frac{1}{\sqrt{6}}(1, 2, -1), q_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$$

N8 Solution:

See U3-13.5 for the complete solution.

Orthogonal basis: $\{(1, 1, 1, 1), (-2, -1, 1, 2), (16, -17, -13, 14)\}$

N9 Solution:

(a) $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

$$\det(A - \lambda I) = (3 - \lambda)^2 - 1 = 0 \implies \lambda = 4, 2$$

For $\lambda = 4$: $x_1 = (1, 1)$. For $\lambda = 2$: $x_2 = (1, -1)$.

(b) $B = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

$$\det(B - \lambda I) = \lambda(\lambda + 3) + 2 = \lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0$$

$$\lambda_1 = -1, \lambda_2 = -2$$

For $\lambda = -1$: $(B + I)x = 0 \implies \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} x = 0 \implies x_1 = (1, -1)$

For $\lambda = -2$: $(B + 2I)x = 0 \implies \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} x = 0 \implies x_2 = (1, -2)$

(c) $C = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 2 & 4 \end{bmatrix}$ (lower triangular)

Eigenvalues = diagonal entries: $\lambda_1 = 1, \lambda_2 = 3, \lambda_3 = 4$

For $\lambda_1 = 1$: $(C - I)x = 0 \implies x_1 = (1, 0, 0)^T$ (after row reduction... actually $x_1 = (1, -1, 1/2)^T$ — solve the system)

$$(C - I) = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & 2 & 3 \end{bmatrix}, \text{ row reduce: } x_2 = -x_1, x_1 + 2x_2 + 3x_3 = 0 \implies x_3 = x_1/3$$

Eigenvector: $(3, -3, 1)^T$

$$\text{For } \lambda_2 = 3: (C - 3I) = \begin{bmatrix} -2 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \text{ so } x_1 = 0, 2x_2 + x_3 = 0$$

Eigenvector: $(0, 1, -2)^T$

$$\text{For } \lambda_3 = 4: (C - 4I) = \begin{bmatrix} -3 & 0 & 0 \\ 2 & -1 & 0 \\ 1 & 2 & 0 \end{bmatrix}, \text{ so } x_1 = 0, -x_2 = 0, x_3 \text{ free}$$

Eigenvector: $(0, 0, 1)^T$

N10 Solution:

$$\text{(a) } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Characteristic polynomial: $p(\lambda) = \lambda^2 - 5\lambda - 2$

By Cayley–Hamilton: $A^2 - 5A - 2I = 0$

$$\text{Multiply by } A^{-1}: A - 5I - 2A^{-1} = 0 \implies A^{-1} = \frac{1}{2}(A - 5I) = \frac{1}{2} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

$$\text{Verify: } AA^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

$$\text{(b)} A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

$$p(\lambda) = (2 - \lambda)(3 - \lambda) = \lambda^2 - 5\lambda + 6$$

$$A^2 - 5A + 6I = 0 \implies A^2 = 5A - 6I$$

$$A^3 = A \cdot A^2 = A(5A - 6I) = 5A^2 - 6A = 5(5A - 6I) - 6A = 19A - 30I$$

$$A^3 = 19 \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} - 30 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 19 \\ 0 & 27 \end{bmatrix}$$

N11 Solution:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$$

$$\lambda_1 = 5, \lambda_2 = 2$$

$$\text{For } \lambda_1 = 5: (A - 5I)x = 0 \implies \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} x = 0 \implies x_1 = (1, 1)^T$$

$$\text{For } \lambda_2 = 2: (A - 2I)x = 0 \implies \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} x = 0 \implies x_2 = (1, -2)^T$$

$$S = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\det(S) = -3, S^{-1} = \frac{1}{-3} \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A^5 = S\Lambda^5 S^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 5^5 & 0 \\ 0 & 2^5 \end{bmatrix} \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 3125 & 32 \\ 3125 & -64 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 6282 & 3093 \\ 6186 & 3189 \end{bmatrix} = \begin{bmatrix} 2094 & 1031 \\ 2062 & 1063 \end{bmatrix}$$

N12 Solution:

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

Eigenvalues: $\lambda_1 = 9, \lambda_2 = 1$ (using $a \pm b$ formula for $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$)

Eigenvectors: $q_1 = \frac{1}{\sqrt{2}}(1, 1)^T, q_2 = \frac{1}{\sqrt{2}}(1, -1)^T$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \Lambda = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = Q\Lambda Q^T \checkmark$$

N13 Solution:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, W = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A^T W A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 9 \end{bmatrix}$$

(Check: $r_{11} = 2 + 1 + 2 = 5, r_{12} = 0 + 1 + 4 = 5, r_{21} = 0 + 1 + 4 = 5, r_{22} = 0 + 1 + 8 = 9$)

$$A^T W b = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 + 3 + 8 \\ 0 + 3 + 16 \end{bmatrix} = \begin{bmatrix} 13 \\ 19 \end{bmatrix}$$

$$\det = 45 - 25 = 20$$

$$\hat{x} = \frac{1}{20} \begin{bmatrix} 9 & -5 \\ -5 & 5 \end{bmatrix} \begin{bmatrix} 13 \\ 19 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 117 - 95 \\ -65 + 95 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 22 \\ 30 \end{bmatrix} = \begin{bmatrix} 1.1 \\ 1.5 \end{bmatrix}$$

Best fit line (weighted): $y = 1.1 + 1.5t$

N14 Solution:

$$(a) f = 3x_1^2 + 2x_2^2 - 4x_1x_2: a_{11} = 3, a_{22} = 2, 2a_{12} = -4 \implies a_{12} = -2$$

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix}$$

$$(b) g = x_1^2 + 4x_2^2 + x_3^2 + 2x_1x_2 - 6x_1x_3 + 4x_2x_3$$

$$a_{11} = 1, a_{22} = 4, a_{33} = 1; 2a_{12} = 2 \implies a_{12} = 1; 2a_{13} = -6 \implies a_{13} = -3; \\ 2a_{23} = 4 \implies a_{23} = 2$$

$$A = \begin{bmatrix} 1 & 1 & -3 \\ 1 & 4 & 2 \\ -3 & 2 & 1 \end{bmatrix}$$

N15 Solution:

$$(a) A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\text{Eigenvalues: } \lambda^2 - 4\lambda + 3 = 0 \implies \lambda = 3, 1 \text{ — both positive} \rightarrow \mathbf{PD}$$

$$\text{Minors: } \det(A_1) = 2 > 0, \det(A) = 4 - 1 = 3 > 0 \rightarrow \mathbf{PD} \checkmark$$

$$\text{Pivots: } d_1 = 2, d_2 = 2 - 1/2 = 3/2 \text{ — both positive} \rightarrow \mathbf{PD} \checkmark$$

$$(b) B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

$$\det(B) = 3 - 4 = -1 < 0 \rightarrow \mathbf{Indefinite}$$

$$(c) C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\text{Minors: } \det(C_1) = 2 > 0, \det(C_2) = 4 - 1 = 3 > 0, \det(C_3) = 2(4 - 1) - 1(2 - 0) + 0 = 6 - 2 = 4 > 0 \rightarrow \mathbf{PD}$$

N16 Solution:

$$(a) A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

$$d_1 = 4, l_{21} = 2/4 = 1/2, d_2 = 3 - (1/2)^2 \cdot 4 = 2$$

$$R = \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{bmatrix}$$

$$\text{Verify: } R^T R = \begin{bmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} \checkmark$$

$$(b) A = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}$$

$$d_1 = 9, l_{21} = 3/9 = 1/3, d_2 = 5 - (1/3)^2 \cdot 9 = 5 - 1 = 4$$

$$R = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\text{Verify: } R^T R = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix} \checkmark$$

N17 Solution:

$$(a) A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \text{ eigenvalues: } \lambda_1 = 3, \lambda_2 = 1$$

$$\sigma_1 = \sqrt{3}, \sigma_2 = 1$$

$$v_1 = \frac{1}{\sqrt{2}}(1, 1)^T, v_2 = \frac{1}{\sqrt{2}}(1, -1)^T$$

$$u_1 = \frac{Av_1}{\sqrt{3}} = \frac{1}{\sqrt{6}}(2, 1, 1)^T$$

$$u_2 = \frac{Av_2}{1} = \frac{1}{\sqrt{2}}(0, 1, -1)^T$$

$u_3 = \text{orthogonal to } u_1, u_2: u_3 = \frac{1}{\sqrt{3}}(1, -1, -1)^T$ (verify)

$$\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

(b) $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ (diagonal, already in SVD form)

$$U = I_2, \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, V = I_2$$

(c) $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$$AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \text{ eigenvalues: } 3, 1$$

$$\sigma_1 = \sqrt{3}, \sigma_2 = 1$$

$$u_1 = \frac{1}{\sqrt{2}}(1, 1)^T, u_2 = \frac{1}{\sqrt{2}}(1, -1)^T$$

$$v_1 = \frac{A^T u_1}{\sqrt{3}} = \frac{1}{\sqrt{6}}(1, 1, 2)^T$$

$$v_2 = \frac{A^T u_2}{1} = \frac{1}{\sqrt{2}}(1, -1, 0)^T$$

$$v_3 = \text{orthogonal to } v_1, v_2: v_3 = \frac{1}{\sqrt{3}}(-1, -1, 1)^T$$

$$\Sigma = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

N18 Solution:

$$X = \begin{bmatrix} 80 & 85 \\ 70 & 75 \\ 60 & 65 \\ 88 & 82 \\ 72 & 74 \end{bmatrix}$$

(a) Column means: $\mu_1 = 74, \mu_2 = 76.2$

$$\tilde{X} = \begin{bmatrix} 6 & 8.8 \\ -4 & -1.2 \\ -14 & -11.2 \\ 14 & 5.8 \\ -2 & -2.2 \end{bmatrix}$$

(b) $\tilde{X}^T \tilde{X} = \begin{bmatrix} 448 & 300.8 \\ 300.8 & 242.8 \end{bmatrix}$ (approximately)

$$C = \frac{1}{4} \tilde{X}^T \tilde{X} = \begin{bmatrix} 112 & 75.2 \\ 75.2 & 60.7 \end{bmatrix}$$

(c) $\det(C - \lambda I) = (112 - \lambda)(60.7 - \lambda) - 75.2^2 = 0$

$$\lambda^2 - 172.7\lambda + (112)(60.7) - 5655 = 0$$

$$\lambda^2 - 172.7\lambda + 1143.4 = 0$$

$$\lambda_1 \approx 165.9, \lambda_2 \approx 6.8$$

(d) Total variance = 172.7

PC1 explains $165.9/172.7 \approx 96.1\%$, PC2 explains $\approx 3.9\%$

(The two subjects are highly correlated, so PC1 captures almost all variance.)

N19 Solution:

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

(a) Column means: $\mu = (3, 4)$

$$\tilde{X} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

$$(b) \tilde{X}^T \tilde{X} = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$$

Eigenvalues: 16 and 0.

$$\sigma_1 = 4, \sigma_2 = 0.$$

$$v_1 = \frac{1}{\sqrt{2}}(1, 1)^T \text{ (eigenvector for } \lambda = 16)$$

$$(c) \text{ First principal component direction: } w_1 = \frac{1}{\sqrt{2}}(1, 1)^T$$

This means the data varies equally in both feature directions (features 1 and 2 are perfectly correlated after centering).

N20 Solution:

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$(a) \det(A - \lambda I) = (4 - \lambda)(1 - \lambda) - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5) = 0$$

$$\lambda_1 = 5, \lambda_2 = 0$$

$$(b) R(x)_{\max} = \lambda_{\max} = 5, R(x)_{\min} = \lambda_{\min} = 0$$

$$(c) \text{ Maximum at eigenvector for } \lambda = 5: (A - 5I)x = 0 \implies -x_1 + 2x_2 = 0 \implies x_1 = (2, 1)^T$$

$$\text{Unit vector: } q_1 = \frac{1}{\sqrt{5}}(2, 1)^T$$

$$\text{Minimum at eigenvector for } \lambda = 0: Ax = 0 \implies 4x_1 + 2x_2 = 0 \implies x_2 = (1, -2)^T$$

$$\text{Unit vector: } q_2 = \frac{1}{\sqrt{5}}(1, -2)^T$$

MASTER REVISION — Final Cheat Sheet

UNIT 3 — Complete Summary

Core Definitions

Term	Definition
Dot product	$x^T y = \sum x_i y_i$
Norm	$ x = \sqrt{x^T x}$
Orthogonal vectors	$x^T y = 0$
Orthogonal complement	$V^\perp = \{x : x^T v = 0 \ \forall v \in V\}$
Projection onto line	$p = \frac{a^T b}{a^T a} a$
Projection matrix (line)	$P = \frac{a a^T}{a^T a}$
Projection onto $C(A)$	$P = A(A^T A)^{-1} A^T$
Normal equations	$A^T A \hat{x} = A^T b$
Orthonormal	$q_i^T q_j = \delta_{ij}$
Orthogonal matrix	$Q^T Q = I, Q^{-1} = Q^T$
Eigenvalue equation	$Ax = \lambda x$
Characteristic equation	$\det(A - \lambda I) = 0$
Cayley–Hamilton	$p(A) = 0$
Diagonalization	$A = S \Lambda S^{-1}$

Key Theorems

1. **Orthogonal $\Rightarrow$ Independent:** Mutually orthogonal nonzero vectors are linearly independent.
2. **Fundamental Theorem:** $C(A) \perp N(A^T)$ and $C(A^T) \perp N(A)$.
3. **Projection properties:** $P^T = P$ and $P^2 = P$.
4. **Symmetric matrices:** Real eigenvalues; orthogonal eigenvectors for distinct eigenvalues.
5. **Gram–Schmidt:** Converts independent vectors to orthonormal basis.
6. **QR:** $A = QR$ with Q orthonormal columns, R upper triangular.

Gram–Schmidt Quick Reference

$$u_1 = v_1$$

$$u_k = v_k - \sum_{j=1}^{k-1} \frac{u_j^T v_k}{u_j^T u_j} u_j$$

$$q_i = \frac{u_i}{\|u_i\|}$$

Eigenvalue Shortcuts

- Triangular matrix: eigenvalues = diagonal entries
 - $\text{tr}(A) = \sum \lambda_i$
 - $\det(A) = \prod \lambda_i$
 - A^{-1} has eigenvalues $1/\lambda_i$
 - A^k has eigenvalues λ_i^k
-

UNIT 4 — Complete Summary

Core Definitions

Term	Definition
Quadratic form	$f(x) = x^T A x$
PD	$x^T A x > 0$ for all $x \neq 0$
PSD	$x^T A x \geq 0$ for all x
ND	$x^T A x < 0$ for all $x \neq 0$
Indefinite	Takes both signs
Cholesky	$A = R^T R$ (PD matrices)
Singular values	$\sigma_i = \sqrt{\lambda_i(A^T A)}$
SVD	$A = U \Sigma V^T$
PCA	Directions of max variance
Rayleigh quotient	$R(x) = x^T A x / x^T x$

PD Tests (All Equivalent for Symmetric A)

PD $\iff$ all eigenvalues > 0 $\iff$ all leading minors > 0 $\iff$ all pivots > 0

SVD Algorithm

1. Compute $A^T A$ (or $A A^T$ if $m < n$)
2. Find eigenvalues $\lambda_i \rightarrow$ singular values $\sigma_i = \sqrt{\lambda_i}$
3. Eigenvectors of $A^T A \rightarrow$ columns of V
4. $u_i = A v_i / \sigma_i \rightarrow$ columns of U
5. Complete U to orthonormal basis if needed

PCA Algorithm

1. Center data: $\tilde{X} = X - \text{mean}$
 2. Covariance: $C = \frac{1}{n-1} \tilde{X}^T \tilde{X}$ (or $\tilde{X} \tilde{X}^T$)
 3. Eigendecompose C : $Cq_i = \lambda_i q_i$
 4. PCs = eigenvectors sorted by λ_i (descending)
 5. Explained variance = $\lambda_k / \sum \lambda_i$
-

Common Traps and Mistakes

Unit 3 Traps

Mistake	Correct Approach
Using $a^T b / a^T b$ in projection	Denominator is $a^T a$, not $a^T b$
Forgetting to verify $q \perp a$	Always check $a^T q = 0$
Skipping a projection in Gram–Schmidt	Subtract projections onto ALL previous u_j
Using v_j instead of u_j in Gram–Schmidt	Always project onto the already-orthogonalized vectors
Confusing $P^2 = P$ with $P^2 = I$	Projection matrices are idempotent, not involutions
Forgetting to normalize in Gram–Schmidt	Orthogonal $\neq$ orthonormal
Eigenvalues of A^{-1}	They are $1/\lambda_i$, not $-\lambda_i$

Unit 4 Traps

Mistake	Correct Approach
Cross-term coefficient	Coefficient of $x_i x_j$ in $x^T A x$ is $2a_{ij}$, not a_{ij}

Mistake	Correct Approach
PD test: only checking $\det > 0$	Need ALL leading principal minors > 0
Forgetting to center data in PCA	Always subtract mean before computing covariance
Row vs column mean in PCA	Features $<$ samples $\rightarrow$ row mean; features $>$ samples $\rightarrow$ column mean
SVD: $u_i = Av_i$ (missing σ_i)	$u_i = Av_i/\sigma_i$ (must divide by singular value)
Confusing $A^T A$ and AA^T	$A^T A$ gives V ; AA^T gives U
Singular values of diagonal matrix	For diagonal A , singular values = $ a_{ii} $

Quick Strategy for Exam Questions

"Find eigenvalues and eigenvectors"

1. Write $\det(A - \lambda I) = 0$
2. Expand and solve for λ
3. For each λ : solve $(A - \lambda I)x = 0$
4. Check: $\text{tr}(A) = \sum \lambda_i$ and $\det(A) = \prod \lambda_i$

"Find the projection of b onto $C(A)$ "

1. Compute $A^T A$ and $A^T b$
2. Solve $A^T A \hat{x} = A^T b$
3. Compute $p = A \hat{x}$
4. Compute $q = b - p$
5. Verify $A^T q = 0$

"Apply Gram–Schmidt"

1. Set $u_1 = v_1$
2. For each new vector: subtract all previous projections
3. Normalize at the end
4. Verify orthogonality by checking dot products

"Find SVD"

1. Compute $A^T A$ (or AA^T)
2. Find eigenvalues $\rightarrow$ singular values
3. Find eigenvectors $\rightarrow V$ (or U)
4. Compute the other set of singular vectors using $u_i = Av_i/\sigma_i$
5. Write $A = U\Sigma V^T$

"Test for positive definiteness"

- **Fastest:** Check leading principal minors (2×2 : $a > 0$ and $\det > 0$)
- **Most reliable:** Find eigenvalues
- **For Cholesky:** Use pivot method

"PCA"

1. Identify if rows or columns are features
2. Compute appropriate mean and center
3. Compute covariance matrix
4. Find eigenvalues and eigenvectors
5. Report explained variance

Important Formulas at a Glance

$$\text{proj}_a(b) = \frac{a^T b}{a^T a} a \quad P = \frac{aa^T}{a^T a} \quad P = A(A^T A)^{-1} A^T$$
$$A^T A \hat{x} = A^T b \quad A = QR \quad A = S\Lambda S^{-1} \quad A = Q\Lambda Q^T$$

$$A = U\Sigma V^T \quad \sigma_i = \sqrt{\lambda_i(A^T A)} \quad u_i = \frac{Av_i}{\sigma_i}$$

$$C = \frac{1}{n-1} \tilde{X}^T \tilde{X} \quad R(x) = \frac{x^T A x}{x^T x} \quad \lambda_{\min} \leq R(x) \leq \lambda_{\max}$$

$$\text{PD} \iff \lambda_i > 0 \iff \text{minors} > 0 \iff \text{pivots} > 0 \quad A = R^T R \text{ (Cholesky)}$$

Concepts Checklist

Unit 3

- ☐ Dot product, norm, angle, orthogonality condition
- ☐ Orthogonal complement and dimension formula
- ☐ Four fundamental subspaces and their orthogonality
- ☐ Projection onto a line (formula and derivation)
- ☐ Rank-1 projection matrix and its properties
- ☐ Projection onto column space (normal equations)
- ☐ Least squares — geometric meaning and normal equations
- ☐ Line fitting via least squares
- ☐ Weighted least squares
- ☐ Orthonormal vectors and orthogonal matrices
- ☐ Gram–Schmidt process (3 or 4 vectors)
- ☐ QR factorization idea
- ☐ Eigenvalues and eigenvectors (definition and computation)
- ☐ Properties of eigenvalues (trace, determinant, shortcuts)
- ☐ Cayley–Hamilton theorem (statement and application)
- ☐ Diagonalization ($A = S\Lambda S^{-1}$)
- ☐ Orthogonal diagonalization of symmetric matrices

Unit 4

- ☐ Quadratic forms (read matrix from form and vice versa)
- ☐ Definiteness definitions (PD, PSD, ND, NSD, Indefinite)
- ☐ Three tests for positive definiteness

- ☐ Proof that A PD $\implies A^{-1}$ PD
 - ☐ Cholesky decomposition ($A = R^T R$)
 - ☐ LDL^T factorization
 - ☐ Singular values (definition and computation)
 - ☐ SVD ($A = U\Sigma V^T$) — full algorithm
 - ☐ Low-rank approximation idea
 - ☐ PCA — centering, covariance, principal components
 - ☐ PCA via SVD vs via covariance matrix
 - ☐ Explained variance ratio
 - ☐ Rayleigh quotient and min/max principles
-

End of Complete Study Guide

Reference: Gilbert Strang, "Linear Algebra and Its Applications", 5th Edition

Course: UE24MA241B — Linear Algebra and Its Applications

⚡ EXAM BOOSTER SECTIONS (NEW)

⚡ MCQ Concept Booster — Unit 3

One-line facts for rapid recall. Read these the night before the exam.

Dot Product & Orthogonality

- Orthogonal vectors $\rightarrow$ dot product = 0
- $\|x\|^2 = x^T x$ (not xx^T , which is a matrix)
- $\cos \theta = x^T y / (\|x\| \|y\|) \rightarrow$ orthogonal means $\theta = 90^\circ$
- Mutually orthogonal nonzero vectors $\rightarrow$ **always linearly independent**
- $(Ax)^T y = x^T (A^T y)$ — transpose moves across dot product

Orthogonal Complement

- $V^\perp$ = all vectors perpendicular to every vector in V
- $\dim(V) + \dim(V^\perp) = n$ (ambient dimension)
- $(V^\perp)^\perp = V$ always
- $V \cap V^\perp = \{0\}$ always

Four Subspaces

- $C(A) \perp N(A^T)$ — column space $\perp$ left null space (both in $\mathbb{R}^m$)
- $C(A^T) \perp N(A)$ — row space $\perp$ null space (both in $\mathbb{R}^n$)
- $N(A^T)$ = left null space (solutions to $A^T y = 0$)

Projections

- Projection onto line $\text{span}(a)$: $p = \frac{a^T b}{a^T a} a$ — denominator is $a^T a$, NOT $a^T b$
- Rank-1 projection matrix: $P = \frac{a a^T}{a^T a}$ — outer product divided by scalar
- Projection onto $C(A)$: $P = A(A^T A)^{-1} A^T$
- Projection matrix properties: $P^T = P$ (symmetric) AND $P^2 = P$ (idempotent)
- Eigenvalues of any projection matrix: only 0 or 1
- Residual $q = b - p$ always lies in $N(A^T)$

Least Squares

- Normal equations: $A^T A \hat{x} = A^T b$ (not AA^T , not $A^T A x = b$)
- Residual $e = b - A \hat{x}$ is **orthogonal to $C(A)$**
- $A^T A$ is invertible iff A has independent columns
- Weighted: $A^T W A \hat{x} = A^T W b$ where $W = \text{diag}(w_i)$

Orthogonal Matrices

- $Q^T Q = I \rightarrow Q^{-1} = Q^T$ (for square Q)
- $\|Qx\| = \|x\|$ — orthogonal matrices preserve length
- $\det(Q) = \pm 1$ always
- For rectangular Q with orthonormal columns: $Q^T Q = I$ but $Q Q^T \neq I$

Gram–Schmidt

- Input: independent vectors; Output: orthonormal vectors spanning same space
- Each step: subtract ALL previous projections, not just the last
- Orthogonal $\neq$ orthonormal (must normalize separately)
- QR: $A = QR$ where Q has orthonormal columns, R is upper triangular

Eigenvalues

- $Ax = \lambda x$ with $x \neq 0 \rightarrow x$ is eigenvector, λ is eigenvalue
- Characteristic equation: $\det(A - \lambda I) = 0$
- Triangular matrix $\rightarrow$ eigenvalues = diagonal entries (fastest shortcut)
- $\text{tr}(A) = \sum \lambda_i$, $\det(A) = \prod \lambda_i$
- Symmetric matrix $\rightarrow$ eigenvalues are real, eigenvectors for different λ are orthogonal
- A singular $\iff \lambda = 0$ is an eigenvalue
- Eigenvalues of A^{-1} : $1/\lambda_i$ (same eigenvectors)
- Eigenvalues of A^k : λ_i^k (same eigenvectors)
- Eigenvalues of $A - cI$: $\lambda_i - c$ (same eigenvectors)

Cayley–Hamilton

- Every matrix satisfies its own characteristic polynomial: $p(A) = 0$
- Use to find A^{-1} : multiply $p(A) = 0$ by A^{-1}
- Use to find A^k : reduce using $p(A) = 0$ repeatedly

Diagonalization

- $A = S\Lambda S^{-1}$ where columns of S are eigenvectors
 - $A^k = S\Lambda^k S^{-1}$ — powers become trivial
 - Diagonalizable iff n linearly independent eigenvectors exist
 - Symmetric: $A = Q\Lambda Q^T$ with Q orthogonal (always diagonalizable)
 - Distinct eigenvalues $\rightarrow$ always diagonalizable
-

⚡ MCQ Concept Booster — Unit 4

Quadratic Forms

- $f(x) = x^T A x$ where A is symmetric
- Coefficient of x_i^2 in $f \rightarrow a_{ii}$
- Coefficient of $x_i x_j$ in $f \rightarrow 2a_{ij}$ (the 2 is critical — most common mistake)
- To go from f to A : halve the cross-term coefficients

Definiteness

- PD: $x^T A x > 0$ for all $x \neq 0 \rightarrow$ "bowl shape"
- PSD: $x^T A x \geq 0 \rightarrow$ "flat bowl, touches zero"
- ND: $x^T A x < 0$ for all $x \neq 0 \rightarrow$ "inverted bowl"
- Indefinite: takes both signs $\rightarrow$ "saddle shape"
- PD $\iff$ all eigenvalues $> 0 \iff$ all leading minors $> 0 \iff$ all pivots > 0
- $\det > 0$ alone is NOT sufficient for PD (need ALL leading minors)
- PD matrix $\rightarrow A^{-1}$ is also PD
- $A^T A$ is always PSD; PD iff A has independent columns

Cholesky

- $A = R^T R$ (upper triangular R , positive diagonal)
- Exists iff A is symmetric PD
- First diagonal entry: $r_{11} = \sqrt{a_{11}}$
- Via LDL^T : $R = \sqrt{D} L^T$

Singular Values

- $\sigma_i = \sqrt{\lambda_i(A^T A)}$ — always nonneg
- $A^T A$ and $A A^T$ have same nonzero eigenvalues
- Number of nonzero singular values = rank of A
- For diagonal A : singular values = |diagonal entries|
- For symmetric PD A : singular values = eigenvalues

SVD

- $A = U\Sigma V^T$ — U orthogonal, Σ diagonal, V orthogonal
- V comes from eigenvectors of $A^T A$; U comes from $u_i = Av_i/\sigma_i$
- If $m < n$: compute AA^T first (smaller matrix)
- If $m > n$: compute $A^T A$ first (smaller matrix)
- $A = \sum_{i=1}^r \sigma_i u_i v_i^T$ — outer product expansion
- Best rank- k approx: keep first k terms (largest σ_i)

PCA

- Always center data first (subtract mean)
- Rows < Columns (features < samples) → subtract row means
- Rows > Columns (samples > features) → subtract column means
- Covariance: $C = \frac{1}{n-1} \tilde{X}^T \tilde{X}$ (rows=samples) or $\frac{1}{n-1} \tilde{X} \tilde{X}^T$ (rows=features)
- PC1 = eigenvector of C for largest eigenvalue
- Explained variance of PC $k = \lambda_k / \sum \lambda_i$
- PCA via SVD: columns of V are principal directions

Rayleigh Quotient

- $R(x) = x^T A x / x^T x$ for symmetric A
 - $\lambda_{\min} \leq R(x) \leq \lambda_{\max}$ always
 - A is PD $\iff R(x) > 0$ for all $x \neq 0 \iff \lambda_{\min} > 0$
 - PCA maximizes $w^T C w$ subject to $\|w\| = 1 \rightarrow$ solution is eigenvector for $\lambda_{\max}$
-

⚡ Fast Methods & Shortcuts

FAST: Eigenvalues of 2×2 Matrix

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\lambda = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4(ad-bc)}}{2} = \frac{\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det}}{2}$$

Even faster for $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ (symmetric with equal diagonal):

$$\lambda_1 = a + b, \quad \lambda_2 = a - b$$

Example: $\begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \rightarrow \lambda_1 = 8, \lambda_2 = 2$ (instant!)

FAST: Projection onto a Line (3 steps)

1. Compute $a^T b$ (dot product)
2. Compute $a^T a$ (dot product of a with itself)
3. $p = \frac{a^T b}{a^T a} \cdot a$

Never forget: denominator is $a^T a$, not $a^T b$.

FAST: Definiteness Check for 2×2

For $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$:

Check	Result
$a > 0$ and $ac - b^2 > 0$	PD
$a < 0$ and $ac - b^2 > 0$	ND
$ac - b^2 < 0$	Indefinite
$ac - b^2 = 0$ (and $a, c \geq 0$)	PSD

Fastest check: compute $\det = ac - b^2$ first. If negative $\rightarrow$ indefinite (done!).

FAST: Cholesky for 2×2

For $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ (PD):

$$R = \begin{bmatrix} \sqrt{a} & b/\sqrt{a} \\ 0 & \sqrt{c - b^2/a} \end{bmatrix}$$

Example: $A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}; R = \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{bmatrix}$

FAST: SVD when A is Symmetric PD

If A is symmetric PD with eigendecomposition $A = Q\Lambda Q^T$:

$$\text{SVD is: } U = V = Q, \quad \Sigma = \Lambda$$

No extra computation needed!

FAST: Gram–Schmidt Structured Steps

Template (always follow this order):

```
u1 = v1
u2 = v2 - (u1·v2)/(u1·u1) * u1
u3 = v3 - (u1·v3)/(u1·u1) * u1 - (u2·v3)/(u2·u2) * u2
...
qi = ui / ||ui||
```

Key check after each step: verify $u_k \cdot u_j = 0$ for all $j < k$.

Exam shortcut: If asked for orthogonal basis only (not orthonormal), skip the normalization step.

FAST: Normal Equations for Line Fitting

For $y = C + Dt$ with m data points:

$$\begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum t_i y_i \end{bmatrix}$$

Compute these 5 sums: $m, \sum t, \sum t^2, \sum y, \sum ty$ — then plug in.

FAST: Cayley–Hamilton to Find A^{-1} (2×2)

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $p(\lambda) = \lambda^2 - (a + d)\lambda + (ad - bc)$:

$$A^2 - \text{tr}(A) \cdot A + \det(A) \cdot I = 0$$

Multiply by A^{-1} :

$$A^{-1} = \frac{1}{\det(A)} (\text{tr}(A) \cdot I - A) = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

This is just the standard 2×2 inverse formula — Cayley–Hamilton gives it automatically!

🔥 Important Practice Questions (Exam Level)

Positive Definiteness — All 3 Methods

Q1. Test $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ for positive definiteness using all three methods.

Solution:

Method 1 — Eigenvalues:

$$\det(A - \lambda I) = -\lambda^3 + 6\lambda^2 - 10\lambda + 4$$

Eigenvalues: $\lambda = 2 - \sqrt{2}, 2, 2 + \sqrt{2}$ (all positive) $\rightarrow$ **PD**

Method 2 — Leading Principal Minors:

$$\det(A_1) = 2 > 0$$

$$\det(A_2) = \det \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = 4 - 1 = 3 > 0$$

$$\det(A_3) = 2(4 - 1) - (-1)(-2 - 0) + 0 = 6 - 2 = 4 > 0$$

All positive $\rightarrow$ **PD** ✓

Method 3 — Pivots:

$$\text{Pivot 1: } d_1 = 2$$

$$\text{After eliminating column 1: } \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$\text{Pivot 2: } d_2 = 3/2$$

$$\text{After eliminating column 2: } \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & 0 & 4/3 \end{bmatrix}$$

$$\text{Pivot 3: } d_3 = 4/3$$

All pivots positive $\rightarrow$ **PD** ✓

Q2. Test $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ for positive definiteness.

Solution (Minors method — fastest):

$$\det(A_1) = 1 > 0$$

$$\det(A_2) = 5 - 4 = 1 > 0$$

$$\det(A_3) = 1(15 - 1) - 2(6 - 0) + 0 = 14 - 12 = 2 > 0$$

All positive $\rightarrow$ **PD**

SVD Problems (Exam Level)

Q3. Find the SVD of $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$.

Solution:

$$A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \text{ (already diagonal)}$$

Eigenvalues: $\lambda_1 = 4, \lambda_2 = 1$

Singular values: $\sigma_1 = 2, \sigma_2 = 1$

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ (eigenvectors: } (0, 1) \text{ for } \lambda = 4, (1, 0) \text{ for } \lambda = 1)$$

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{2} \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$u_2 = \frac{Av_2}{\sigma_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ (orthogonal to } u_1, u_2)$$

$$U = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Q4. Find the SVD of $A = \begin{bmatrix} 4 & 3 \\ 0 & 0 \end{bmatrix}$.

Solution:

$$A^T A = \begin{bmatrix} 16 & 12 \\ 12 & 9 \end{bmatrix}$$

Eigenvalues: $\text{tr} = 25, \det = 0 \rightarrow \lambda_1 = 25, \lambda_2 = 0$

$\sigma_1 = 5, \sigma_2 = 0$ (rank 1)

Eigenvector for $\lambda_1 = 25$: $(A^T A - 25I)v = 0$

$$\begin{bmatrix} -9 & 12 \\ 12 & -16 \end{bmatrix} v = 0 \implies 3v_2 = 4v_1 \implies v_1 = \frac{1}{5} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{5} \begin{bmatrix} 4 & 3 \\ 0 & 0 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 25 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (orthogonal to } u_1 \text{)}$$

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I, \quad \Sigma = \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$$

PCA Problems (Basic Level)

Q5. Given $X = \begin{bmatrix} 4 & 2 \\ 3 & 4 \\ 2 & 5 \end{bmatrix}$ (rows = samples, columns = features), find the covariance matrix.

Solution:

Column means: $\mu_1 = (4 + 3 + 2)/3 = 3, \mu_2 = (2 + 4 + 5)/3 = 11/3$

$$\tilde{X} = \begin{bmatrix} 1 & -5/3 \\ 0 & 1/3 \\ -1 & 4/3 \end{bmatrix}$$

$$\tilde{X}^T \tilde{X} = \begin{bmatrix} 1+0+1 & -5/3+0-4/3 \\ -5/3+0-4/3 & 25/9+1/9+16/9 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -3 & 42/9 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -3 & 14/3 \end{bmatrix}$$

$$C = \frac{1}{2} \tilde{X}^T \tilde{X} = \begin{bmatrix} 1 & -3/2 \\ -3/2 & 7/3 \end{bmatrix}$$

Q6. For the covariance matrix $C = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$, find the principal components and explained variance.

Solution:

Eigenvalues: $\det(C - \lambda I) = (3 - \lambda)(2 - \lambda) - 1 = \lambda^2 - 5\lambda + 5 = 0$

$$\lambda_{1,2} = \frac{5 \pm \sqrt{5}}{2}, \text{ so } \lambda_1 \approx 3.618, \lambda_2 \approx 1.382$$

$$\text{Total variance} = \text{tr}(C) = 5$$

$$\text{PC1 explains } \lambda_1/5 = (5 + \sqrt{5})/10 \approx 72.4\%$$

$$\text{PC2 explains } \lambda_2/5 = (5 - \sqrt{5})/10 \approx 27.6\%$$

$$\text{For PC1 direction: } (3 - \lambda_1)v_1 + v_2 = 0 \implies v_2 = (\lambda_1 - 3)v_1 = \frac{\sqrt{5}-1}{2}v_1$$

$$\text{Normalized: } w_1 \approx (0.851, 0.526)$$

Last-Minute Revision (5–10 min)

The 10 Most Important Formulas

$$\boxed{1.} \quad p = \frac{a^T b}{a^T a} \quad \boxed{2.} \quad A^T A \hat{x} = A^T b \quad \boxed{3.} \quad P = A(A^T A)^{-1} A^T$$

$$\boxed{4.} \quad u_k = v_k - \sum_{j < k} \frac{u_j^T v_k}{u_j^T u_j} u_j \quad \boxed{5.} \quad \det(A - \lambda I) = 0$$

$$\boxed{6.} \quad A = S\Lambda S^{-1} \quad \boxed{7.} \quad p(A) = 0 \text{ (Cayley-Hamilton)}$$

$$\boxed{8.} \quad A = U\Sigma V^T \quad \boxed{9.} \quad \sigma_i = \sqrt{\lambda_i(A^T A)} \quad \boxed{10.} \quad C = \frac{1}{n-1} \tilde{X}^T \tilde{X}$$

The 10 Most Common Exam Mistakes

#	Mistake	Correct
1	Projection denominator: $a^T b$	Must be $a^T a$
2	Cross-term in quadratic form: a_{ij}	Must be $2a_{ij}$
3	PD test: only $\det > 0$	Need ALL leading minors > 0
4	SVD: $u_i = Av_i$	Must be $u_i = Av_i / \sigma_i$
5	PCA: skip centering	Always subtract mean first
6	Gram-Schmidt: subtract only last projection	Subtract ALL previous projections
7	$Q^T Q = I$ implies $Q Q^T = I$	Only true for square Q
8	Eigenvalues of A^{-1} : $-\lambda_i$	Correct: $1/\lambda_i$
9	$P^2 = I$ for projection matrix	Correct: $P^2 = P$ (idempotent)
10	PCA row/column mean confusion	Features $<$ samples $\rightarrow$ row mean

Quick Recall Checklist (tick before exam)

Unit 3:

- ☐ $x \perp y \iff x^T y = 0$
- ☐ $\dim(V) + \dim(V^\perp) = n$
- ☐ $C(A) \perp N(A^T), C(A^T) \perp N(A)$
- ☐ Projection formula: $p = \frac{a^T b}{a^T a} a$
- ☐ Normal equations: $A^T A \hat{x} = A^T b$

- ☐ $P^T = P$ and $P^2 = P$
- ☐ Gram–Schmidt: subtract all previous projections
- ☐ $Q^T Q = I \implies Q^{-1} = Q^T$
- ☐ Eigenvalues: $\det(A - \lambda I) = 0$
- ☐ $\text{tr}(A) = \sum \lambda_i$, $\det(A) = \prod \lambda_i$
- ☐ Cayley–Hamilton: $p(A) = 0$
- ☐ Diagonalization: $A = S\Lambda S^{-1}$, $A^k = S\Lambda^k S^{-1}$

Unit 4:

- ☐ Quadratic form: cross-term coefficient = $2a_{ij}$
- ☐ PD iff all eigenvalues > 0 iff all leading minors > 0 iff all pivots > 0
- ☐ Cholesky: $A = R^T R$ (PD only)
- ☐ SVD: $A = U\Sigma V^T$, $\sigma_i = \sqrt{\lambda_i(A^T A)}$
- ☐ $u_i = Av_i / \sigma_i$ (divide by singular value!)
- ☐ PCA: center $\rightarrow$ covariance $\rightarrow$ eigendecompose
- ☐ Explained variance = $\lambda_k / \sum \lambda_i$
- ☐ Rayleigh: $\lambda_{\min} \leq R(x) \leq \lambda_{\max}$

Syllabus Priority Map

Topic	Exam Weight	Focus
Projections + Least Squares	★ ★ ★ ★ ★	Highest — always in exam
Gram–Schmidt	★ ★ ★ ★ ★	Always in exam
Eigenvalues + Diagonalization	★ ★ ★ ★ ★	Always in exam
Cayley–Hamilton	★ ★ ★ ★	Very frequent
Positive Definiteness	★ ★ ★ ★	Very frequent
Quadratic Forms	★ ★ ★	Frequent
SVD	★ ★ ★	Moderate — know the algorithm
Cholesky	★ ★ ★	Moderate

Topic	Exam Weight	Focus
PCA	☆ ☆	Know centering + covariance
Rayleigh Quotient	☆ ☆	Know the formula + bounds

Corrections applied in this revision:

- Section 9.5: Fixed $A^T b$ from $[6, 0]$ to $[6, 2]$; corrected $\hat{x}$ to $[16/7, -3/7]$
- N6 Solution: Fixed $A^T b$ from $[5, 7]$ to $[5, 8]$; corrected $\hat{x}$ to $[1, 1]$
- N13 Solution: Fixed $A^T W A$ from $[[5, 6], [6, 10]]$ to $[[5, 5], [5, 9]]$; corrected $\hat{x}$ to $[1.1, 1.5]$

End of Exam Booster Sections